**PROMPT-1: I am studying the supplied research paper on answerability-aware contract intelligence. Before looking at its technical implementation, I want to understand the research motivation. What specific problem motivated the authors to develop this framework? Why might conventional information extraction systems struggle when a document does not contain the information being requested? Explain the research problem using the paper's own terminology and examples.**

**1. The central research problem**

The motivation behind **answerability-aware contract intelligence** is that extracting information from a contract is not simply about finding an answer. It is also about determining whether the requested information actually exists in the document.

Traditional information extraction systems are generally designed to identify and extract information that is present in a document. However, real-world contracts do not necessarily contain every piece of information a user might ask about.

This creates an important problem: **a system may produce an answer even when the document does not contain sufficient evidence to support that answer.**

In contract analysis, this is particularly problematic because incorrect answers can lead to misunderstandings about contractual obligations, rights, deadlines, and liabilities.

**2. Why conventional information extraction struggles**

The fundamental issue is the difference between two situations:

- **Answerable question:** The contract contains the information needed to answer the question.
- **Unanswerable question:** The contract does not contain the requested information, or does not provide sufficient evidence to establish an answer.

Conventional extraction systems tend to focus on locating an answer span in the document. This creates difficulties when the correct response is that the information is absent.

For example, consider a contract that specifies:

- The commencement date of a lease.
- The monthly rental payment.
- The duration of the lease.

Now suppose someone asks:

*Does the contract specify an option to purchase the leased asset?*

If the contract contains no such clause, the system should identify the question as unanswerable.

A conventional extraction system may nevertheless select a sentence that mentions the asset, lease duration, or another related provision. That sentence might look relevant, but it does not establish the requested fact.

This is the distinction the research addresses: **finding text that appears relevant is not the same as finding evidence that answers the question.**

**3. Why this matters in contract intelligence**

The problem becomes more significant in accounting-oriented contract analysis.

The paper focuses on Right-of-Use (ROU) asset management under **IFRS 16**.

The objective is to identify lease-related contractual evidence and convert it into accounting-oriented indicators.

If a system incorrectly treats missing contractual information as available information, it can produce misleading indicators and unreliable downstream analysis.

The framework therefore emphasizes:

- Reliable identification of contractual evidence.
- Explicit determination of whether a question is answerable.

- Traceability between extracted information and its supporting contractual evidence.
- Conservative handling of missing information.

**4. What the paper actually proposes**

The authors introduce **ROU-TransQA-v2**, which explicitly distinguishes between answerable and unanswerable questions.

The paper uses three important ideas:

| **Concept** | **Meaning** |
|---|---|
| Answerability classification | Determines whether the contract contains sufficient evidence to answer a question. |
| Extractive question answering | Identifies the relevant answer span in the contract when evidence exists. |
| Conservative evidence management | Avoids treating unsupported information as established contractual facts. |

An important detail from the paper is its use of the **CUAD (Contract Understanding Atticus Dataset)** as a legal contract-QA benchmark, rather than as an IFRS 16-specific dataset.

The authors report that 68.20% of their independent test instances are unanswerable. This illustrates why detecting missing

information is not a minor feature: it is a substantial part of the problem.

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**5. The research motivation in one statement**

The research was motivated by a limitation in conventional contract information extraction: these systems focus on extracting information from available text but do not adequately address whether the requested information is actually supported by the contract.

The authors want a system that knows the difference between:

1. Information explicitly supported by contractual evidence.
2. Information that cannot be established because the required evidence is absent.

**The central research question is therefore not merely:**

"What information can we extract from this contract?"

It is also:

"Can this question actually be answered using the evidence available in this contract?"

That distinction is the foundation of answerability-aware contract intelligence.

*Source: An Answerability-Aware Contract Intelligence Framework for Right-of-Use Asset Management With IFRS 16-Oriented Indicator Mapping and Multidimensional Analytics, IEEE Access, 2026.*

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**PROMPT-2: Following the research problem discussed above, I want to understand the concept of answerability more deeply. Why should an information extraction system determine whether evidence exists before producing an answer? What is the difference between an answerable question, an unanswerable question and a question with ambiguous evidence? What could go wrong if the model always attempts to produce an answer? Explain using examples from the supplied paper.**

**1. Why answerability must be checked first**

An information extraction system should establish whether sufficient evidence exists before generating an answer because **an answer must be supported by the document, not merely sound plausible.**

Consider a lease agreement.

A user asks:

"Does the agreement specify the commencement date of the lease?"

There are two possible situations:

- The contract explicitly mentions the commencement date. The system can extract it.
- The contract does not mention the commencement date. The system should not invent or infer one.

The important principle is that absence of evidence must not be converted into an assumed fact.

This matters especially in IFRS 16-related contract analysis, where extracted information may subsequently be used for accounting-oriented indicators.

**2. Three possible evidence situations**

These distinctions are important for understanding answerability-aware extraction.

**A. Answerable question**

Sufficient evidence exists

A question is answerable when the contract contains explicit information that supports the requested answer.

**Example**

Contract clause:

"The lease shall commence on 1 January 2026 and continue for a period of five years."

Question:

"When does the lease commence?"

Expected output:

- Answer: 1 January 2026.
- Evidence: The commencement clause.
- Answerability: Yes.

The system can identify the answer span directly from the contract.

**B. Unanswerable question**

Evidence is absent

A question is unanswerable when the requested information is not available in the contract.

**Example**

Contract clause:

"The lease shall commence on 1 January 2026 and continue for five years."

Question:

"Does the agreement provide an option for the lessee to purchase the asset at the end of the lease?"

Suppose the agreement contains no purchase option clause.

Expected output:

- Answer: Not available in the contract.
- Evidence: No supporting clause exists.
- Answerability: No.

The system must abstain rather than manufacture an answer from unrelated contractual language.

An important distinction: an unanswerable question does not necessarily mean that the underlying fact is false. It means the document cannot establish that fact.

For example, the lessee might have a purchase option in a separate agreement. Its absence from the current contract does not prove that no such arrangement exists elsewhere.

**C. Question with ambiguous evidence**

Evidence is insufficiently clear

This situation is slightly different from complete absence of information.

The contract contains related information, but it does not clearly establish the answer being requested.

**Example**

Contract clause:

"The lessee may extend the lease subject to mutual agreement between the parties."

Question:

"Does the lessee have an unconditional right to extend the lease?"

The contract mentions an extension, but the extension depends on mutual agreement.

Therefore:

- Relevant contractual language exists.
- However, it does not establish an unconditional extension right.
- The system should not interpret the extension clause as proof of an unconditional right.

This is where merely matching keywords becomes dangerous.

The word *extend* appears in both the question and the contract, but the actual contractual meaning is different.

**3. What goes wrong if the model always produces an answer?**

This is the central failure that answerability-aware systems are designed to prevent.

Consider four potential problems.

| **Failure** | **What happens** |
|---|---|
| Hallucination | The model generates information that does not exist in the contract. |
| False extraction | It selects a related sentence that does not actually answer the question. |
| Unsupported inference | It assumes contractual rights or obligations without sufficient evidence. |
| Downstream accounting errors | Incorrect extracted information influences subsequent accounting indicators and analysis. |

**Example: Incorrect lease extension interpretation**

Suppose the contract says:

"The lease may be extended subject to mutual agreement."

The system is asked:

"Does the lessee have an extension option?"

A model that always attempts to answer might return:

"Yes, the lessee has an extension option."

This is potentially misleading because the clause does not establish an unconditional right held by the lessee.

The incorrect answer could then influence the assessment of lease-related indicators.

The problem is not necessarily that the model failed to locate relevant text. It located relevant text but interpreted its evidential significance incorrectly.

**4. How this connects to the paper's framework**

The paper introduces **ROU-TransQA-v2**, which combines two important capabilities:

1. RoBERTa-based extractive question answering.
2. Explicit answerability classification.

It also uses conditional masking of span supervision for unanswerable instances.

The distinction is important:

- For answerable instances, the model learns to identify the appropriate answer span.
- For unanswerable instances, it should learn not to produce an unsupported answer span.

The paper evaluates these capabilities separately through:

- **HasAns F1:** Performance on answerable questions.
- **NoAns F1:** Performance on unanswerable questions.

One particularly interesting result is that 68.20% of its independent test instances are unanswerable. The authors report a NoAns F1 of 99.34 across five independent test runs, compared with a HasAns F1 of 21.54.

This highlights an important research limitation: strong rejection of unanswerable questions does not automatically mean strong extraction of answerable contractual clauses.

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**A terminology clarification:** The paper's abstract describes explicit answerability classification. It does not establish a separate third model class called *ambiguous*. Ambiguous evidence is useful as a conceptual distinction when studying the framework, but should not automatically be treated as a formally implemented third classification label.

**5. The fundamental takeaway**

The three situations can be understood through this simple example:

| **Situation** | **Contract evidence** | **Correct system behaviour** |
|---|---|---|
| Answerable | Explicit commencement date | Extract the date |
| Unanswerable | No purchase option information | Abstain |
| Ambiguous | Extension requires mutual agreement, but question asks about an unconditional right | Avoid an unsupported conclusion |

The fundamental research principle is:

**An intelligent extraction system must know when to answer, when not to answer, and when the available evidence does not justify the requested conclusion.**

This is what separates reliable contract intelligence from a conventional system that simply tries to find an answer for every question.

Source: Original IEEE Access research paper

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**PROMPT-3: Now that I understand the answerability concept, I want to examine how the authors implemented it. Walk me through the research framework step by step. Explain the role of the model architecture, answerability classification, evidence extraction, training approach and evaluation methodology. Why did the authors choose this approach? Which parts are specific to legal contract intelligence?**

**1. Overview of the research framework**

The authors propose **ROU-TransQA-v2**, an application-specific contract intelligence framework designed to identify contractual evidence, determine whether questions are answerable, and map extracted information to IFRS 16-oriented indicators.

The framework is not simply a question-answering model. It combines several stages:

1. Contract and question processing.
2. RoBERTa-based extractive question answering.
3. Explicit answerability classification.
4. Evidence extraction and traceability.
5. D1–D6 indicator mapping.
6. Multidimensional analytics and evaluation.

The central design principle is to separate **finding an answer** from **determining whether a supported answer exists**.

**ROU-TransQA-v2: conceptual workflow**

**Input: Contract + Question**

Lease agreement and a question about its provisions

**RoBERTa-based QA model**

Contextual language understanding and answer prediction

**Answerable**

Extract supported answer span

**Unanswerable**

Reject unsupported answer

**Evidence management**

Preserve the relationship between the question, answer and contractual evidence

**D1–D6 indicator mapping**

Expert-informed IFRS 16-oriented classification

**Multidimensional analytics**

Structured information for accounting-oriented review

Conceptual representation of the framework, not a reproduction of the paper's exact architectural diagram.

**2. Model architecture: Why RoBERTa?**

The authors use **RoBERTa** as the language understanding backbone.

RoBERTa stands for *Robustly Optimized BERT Pretraining Approach*.

It is a transformer-based language model that learns contextual relationships between words.

Why is this useful for contracts?

Contractual language often contains:

- Long sentences.
- Complex grammatical structures.
- Conditional obligations.
- Exceptions and exclusions.
- Legally significant differences between similar expressions.

Consider:

"The lessee may renew the agreement subject to mutual consent."

A simple keyword-based system might identify the word *renew* and conclude that the lessee has a renewal right.

RoBERTa can use the surrounding context to understand that renewal depends on mutual consent.

**Role of the architecture**

The model performs extractive question answering: rather than generating an entirely new answer, it identifies the relevant portion of the supplied contract.

In a conventional extractive QA architecture, the model predicts:

- Start position: Where does the answer begin?
- End position: Where does the answer end?

For example:

Contract: "The lease commences on 1 January 2026."

Question: "What is the commencement date?"

Predicted answer span: 1 January 2026.

The important advantage is that the answer can be traced to text already present in the contract.

The paper's abstract confirms the use of RoBERTa-based extractive QA, although the exact layer dimensions and implementation configuration should be taken from its architecture section rather than assumed from the model family.

**3. Answerability classification**

This is the most important component of the research.

The model must distinguish between:

- Questions for which a valid answer span exists.
- Questions for which no valid answer span exists.

The authors incorporate **explicit answerability classification** into the framework.

Think of it as a decision mechanism that prevents the extraction system from treating every question as answerable.

For example:

| **Question** | **Contract content** | **Expected decision** |
|---|---|---|
| What is the lease commencement date? | Date explicitly mentioned | Answerable |

| | | |
|---|---|---|
| What is the purchase option price? | No purchase option information | Unanswerable |
| What is the renewal period? | Renewal mentioned, but no duration specified | Cannot establish the requested answer |

The last example is important: a relevant clause does not necessarily provide sufficient evidence for the specific question.

**Why combine classification and extraction?**

These two tasks solve different problems.

- Answerability classification determines whether an answer should be provided.
- Span extraction determines where the answer is located.

If the model only performs span extraction, it may select an irrelevant sentence when no correct answer exists.

By incorporating answerability, the authors attempt to make extraction more conservative.

**4. Evidence extraction and traceability**

Once a question is considered answerable, the framework needs to identify the supporting contractual text.

Consider this example:

**Contract clause**

"The lease term shall commence on 1 January 2026 and remain effective for five years."

**Question**

"What is the lease duration?"

**Extracted answer**

Five years.

**Supporting evidence**

The original contractual clause containing the five-year duration.

This relationship is important because the extracted answer should not exist independently of its source.

In contract intelligence, traceability allows a reviewer to:

- Verify the extracted information.
- Return to the original contractual provision.
- Check whether the answer was interpreted correctly.
- Identify unsupported conclusions.

This is especially important when the extracted information is subsequently used for accounting-oriented analysis.

**5. Training approach**

The authors use the CUAD dataset.

Contract Understanding Atticus Dataset

CUAD is a legal contract question-answering benchmark containing questions about contractual provisions.

One important distinction: **CUAD is not an IFRS 16-specific dataset.** The authors use it as a legal contract-QA benchmark for developing and evaluating their answerability-aware approach.

The training process involves several important ideas.

**A. Training on answerable and unanswerable instances**

The model needs examples of both situations.

For answerable instances, it learns to predict the start and end positions of the correct answer.

For unanswerable instances, it must learn that a valid answer span is unavailable.

Without unanswerable training examples, the model may learn to select the closest matching text even when that text does not answer the question.

**B. Conditional masking of span supervision**

Important technical contribution

The abstract specifically identifies **conditional masking of span supervision for unanswerable instances**.

To understand this, consider how span-based QA training works.

For an answerable question, the training data provides the correct answer positions.

For example:

- Contract: "The lease term is five years."
- Question: "What is the lease term?"
- Start position: Position of "five".
- End position: Position of "years".

But what happens when the answer does not exist?

There are no valid answer positions to supervise.

The authors address this through conditional masking: span supervision is masked for unanswerable instances instead of treating an arbitrary text position as a correct answer.

The model can therefore learn answerability without being forced to learn an incorrect answer span.

This is particularly important because the dataset contains a substantial proportion of unanswerable questions.

**C. Validation and configuration selection**

The authors evaluate model configurations on validation data and select their final configuration based on the resulting performance.

The final model is subsequently evaluated on independent test data.

The paper also compares its approach with Saul-7B-Instruct-v1, a legal-domain instruction-tuned language model.

This comparison helps investigate a trade-off:

- A model may be better at extracting actual contractual answers.
- Another model may be better at rejecting questions that cannot be answered.

The authors report that Saul-7B-Instruct-v1 performs better on answerable-clause extraction and macro-QA F1, whereas ROU-TransQA-v2 retains stronger no-answer rejection.

That comparison is significant: the research is not claiming that RoBERTa is universally superior. It is examining the value of explicit answerability handling.

**6. Evaluation methodology**

The authors evaluate the framework at multiple levels rather than relying on one overall accuracy score.

This is important because a system can perform well at rejecting unanswerable questions while performing poorly at extracting actual answers.

**A. Exact Match (EM)**

Exact Match checks whether the predicted answer exactly matches the reference answer.

Example:

- Reference: Five years.
- Prediction: Five years.
- Result: Correct.

If the prediction is "5 years", it may not count as an exact match depending on normalization.

**B. F1 score**

F1 measures the overlap between the predicted answer and the reference answer using precision and recall.

For example:

- Reference: The lease term is five years.
- Prediction: Five years.

The prediction captures the essential answer, even though it does not reproduce the entire reference text.

**C. HasAns F1 and NoAns F1**

These are particularly important for this research.

| **Metric** | **What it evaluates** |
|---|---|
| HasAns F1 | Extraction performance on answerable questions |
| NoAns F1 | Correct rejection of unanswerable questions |
| Overall F1 | Combined answer performance |
| EM | Exact agreement with reference answers |

Reported independent-test results

**99.34**

**NoAns F1**

± 0.12

**21.54**

**HasAns F1**

± 0.38

**74.60**

**Overall F1**

± 0.30

**73.92**

**Exact Match**

± 0.30

Mean ± standard deviation across five independent test runs. Source: IEEE paper abstract.

An important observation:

**NoAns F1 is extremely high, but HasAns F1 is considerably lower.**

This means the model is very effective at rejecting unanswerable questions, but its extraction of answerable contractual clauses remains a weakness.

The authors explicitly acknowledge this limitation.

**D. Why dataset distribution matters**

The independent test set contains:

**68.20%**

Unanswerable instances

This means almost seven out of every ten test questions do not have an answer in the document.

Consequently, overall F1 can give an overly favourable impression of performance if the model is particularly good at rejecting unanswerable questions.

That is why examining HasAns F1 and NoAns F1 separately is essential.

**E. External enterprise validation**

The authors also evaluate the framework using:

- 60 enterprise contracts.
- 2,460 QA instances.

Reported results include:

| **Metric** | **Result** |
|---|---|
| Exact Match | 70.83 |
| Overall F1 | 71.53 |
| HasAns F1 | 19.83 |
| NoAns F1 | 97.76 |
| D1–D6 mapping macro-F1 | 0.816 |

This provides evidence beyond the CUAD benchmark, although the relatively low HasAns F1 remains an important limitation.

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**7. Which components are specific to legal contract intelligence?**

Not every component of this framework is legally specific.

The distinction is important because the underlying answerability approach could be applied to other document-intensive domains.

| **Component** | **General or domain-specific?** |
|---|---|

| | |
|---|---|
| RoBERTa transformer architecture | General NLP |
| Extractive question answering | General NLP |
| Answerability classification | General NLP |
| Conditional masking of span supervision | General ML training approach |
| CUAD dataset | Legal contract-specific |
| Contractual evidence identification | Legal domain-specific application |
| IFRS 16 indicator mapping | Accounting-specific |
| D1–D6 classification | Framework-specific accounting mapping |
| Multidimensional contract analytics | Application-specific |

**The legal and accounting-specific contribution**

The authors extend ordinary question answering into a domain-oriented workflow.

The system is designed to identify contractual information and connect it to accounting-oriented indicators.

For example:

- Lease commencement information.
- Lease duration.
- Renewal-related provisions.
- Purchase options.
- Other relevant contractual conditions.

These are examples of lease-related information that can be investigated through contract questions. They should not be mistaken for the complete definition of the paper's D1–D6 indicators.

The D1–D6 mapping layer uses expert-informed classification to translate extracted contractual evidence into structured indicators.

Importantly, this mapping does not mean the system automatically calculates or determines all IFRS 16 accounting measurements.

The authors explicitly distinguish accounting-oriented review from automated IFRS 16 measurement.

**8. Why did the authors choose this approach?**

There are five underlying reasons.

1. **RoBERTa:** Provides contextual language understanding for complex contractual sentences.

2. **Extractive QA:** Keeps answers grounded in existing contractual text instead of freely generating information.

3. **Answerability classification:** Allows the system to reject questions that lack sufficient contractual evidence.

4. **Conditional masking:** Prevents unanswerable examples from receiving inappropriate answer-span supervision during training.

5. **D1–D6 mapping:** Connects document-level extraction to the actual accounting-oriented application.

The overall objective is to build a system that is not only capable of identifying information but also conservative about what it claims to know.

**9. What you should understand before studying the implementation**

There are three different research contributions to keep separate.

**Contribution 1: NLP methodology**

RoBERTa-based extractive question answering combined with explicit answerability classification.

**Contribution 2: Evidence reliability**

Conditional masking, conservative answer rejection and traceability of extracted contractual information.

**Contribution 3: Domain application**

Expert-informed D1–D6 mapping and multidimensional analytics for IFRS 16-oriented contract review.

The most important distinction is that the first two contributions concern how a question-answering system works, whereas the third concerns what the extracted information is used for.

**My interpretation of the paper's central technical contribution:** It is not simply the use of RoBERTa for contract extraction. It is the integration of answerability-aware extraction with evidence management and accounting-oriented mapping.

There is also a significant limitation worth remembering: the framework demonstrates strong no-answer rejection, but its answerable-clause extraction performance is weak. The paper therefore establishes a useful conservative evidence-management approach, not a fully solved contract-understanding problem.

**Primary source:** An Answerability-Aware Contract Intelligence Framework for Right-of-Use Asset Management With IFRS 16-Oriented Indicator Mapping and Multidimensional Analytics — IEEE Access (2026)

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**PROMPT-4: I want to understand the significance of the experimental results reported in the paper. The authors report different performance for answerable and unanswerable cases. What does this difference reveal about the challenges of information extraction? Why is it important to evaluate both categories separately? What limitations should be considered when interpreting these results? Do not apply the reported metrics to another domain**.

**1. What do the experimental results reveal?**

The most important finding is the large performance gap between answerable and unanswerable questions.

In the paper's independent test evaluation, averaged over five runs:

| Metric | Result |
|---|---|
| HasAns F1 | 21.54 ± 0.38 |
| NoAns F1 | 99.34 ± 0.12 |
| Overall F1 | 74.60 ± 0.30 |
| Exact Match | 73.92 ± 0.30 |

The model performs exceptionally well at recognising when it should not provide an answer, but much less well at extracting the correct answer when one exists.

This reveals that **knowing when information is absent and successfully locating information when it is present are two different technical challenges.**

**A. Why is answerable information difficult to extract?**

An answerable question does not automatically mean the answer is easy to find.

Consider a lease agreement containing:

The lessee may extend the lease term subject to mutual agreement between both parties.

Question: Does the lessee have an unconditional right to extend the lease?

The contract contains the word "extend", but that does not establish an unconditional right.

The model must understand:

- The meaning of the clause.
- The relationship between the parties.
- The condition attached to the extension.
- Whether the clause actually answers the question.

This is more demanding than identifying whether a question has no answer.

Even when a relevant provision exists, the model may extract an incomplete span, select the wrong provision, or misunderstand a condition.

**B. Why can unanswerable questions be easier?**

Consider:

Contract: "The lease term is five years."

Question: "What is the purchase option price?"

There is no purchase option price in the supplied text.

The model needs to reject the question rather than identify an answer span.

The paper's answerability-aware design is intended to make this rejection reliable.

However, the distinction is not always obvious. A contract may contain related information without providing enough evidence to answer the exact question.

That is why unanswerability detection still requires contextual understanding, even if the reported performance is much higher.

**2. Why must both categories be evaluated separately?**

Because an overall score can hide a serious weakness.

The paper reports that **68.20% of independent test instances are unanswerable**.

That means unanswerable questions make up more than two-thirds of the evaluation set.

Imagine a hypothetical system that rejects nearly every question:

- It would perform very well on unanswerable questions.
- It would fail to extract answers from most answerable questions.

Its overall performance could still look respectable because unanswerable questions dominate the dataset.

This is why the authors report separate metrics.

**What each metric tells us**

| Metric | What it helps diagnose |
| --- | --- |

| | |
|---|---|
| HasAns F1 | Whether the system can find and extract actual answers |
| NoAns F1 | Whether the system correctly abstains when answers are absent |
| Overall F1 | How it performs across the combined evaluation set |
| Exact Match | How often its prediction exactly matches the reference |

Separate evaluation exposes two different failure modes:

1. **False rejection:** The answer exists, but the model says it is unavailable.
2. **False extraction:** The answer does not exist, but the model produces an unsupported answer.

Both matter in contractual information extraction.

A reliable system must avoid both errors. High NoAns F1 alone does not establish reliable information extraction.

**3. What do the results tell us about the framework?**

The results suggest that the framework has achieved an important but incomplete objective.

**What the model does well**

No-answer rejection

99.34 F1

**What remains difficult**

Answer extraction

21.54 F1

Independent test results averaged over five runs. The bars compare the reported F1 values, not the relative difficulty of the tasks.

The authors themselves caution that the aggregate F1 score is strongly influenced by conservative rejection of unanswerable questions.

This is a critical interpretation: the overall score should not be presented as evidence that the framework extracts contractual information reliably.

**4. Limitations when interpreting the results**

There are several limitations to keep in mind.

**A. The imbalance between categories**

Since 68.20% of test instances are unanswerable, the overall score is influenced heavily by no-answer performance.

Therefore, the overall F1 of 74.60 does not mean the model has 74.60 F1 on answerable questions.

The HasAns F1 of 21.54 is the more relevant indicator of its answer-extraction weakness.

**B. Strong rejection does not mean strong extraction**

A model can correctly reject missing information without being particularly good at identifying information that exists.

The difference between 99.34 and 21.54 demonstrates precisely this problem.

The framework's answerability component is promising, but the extraction component still requires improvement.

**C. Benchmark-specific evaluation**

The authors use CUAD, a legal contract-QA benchmark.

Its results should be interpreted in the context of that dataset, its contractual questions, its answer annotations and its answerability distribution.

Performance on this benchmark does not automatically establish performance on every type of contract or contractual question.

**D. External validation is useful but limited**

The authors also report enterprise validation using 60 contracts and 2,460 QA instances.

The reported HasAns F1 is 19.83, while NoAns F1 is 97.76.

This reinforces the same pattern seen in the independent test evaluation: rejection is considerably stronger than answer extraction.

However, 60 contracts are not sufficient to establish universal reliability across all contractual settings.

**E. Different evaluation tasks measure different capabilities**

The paper also reports D1–D6 mapping performance:

- Enterprise validation macro-F1: 0.816.
- Expert validation macro-F1: 0.842.
- Expert validation accuracy: 0.862.

These measure indicator-mapping performance, not extractive QA performance.

A strong D1–D6 mapping score should not be interpreted as evidence that the system extracts answerable clauses equally well.

**F. Exact Match has its own limitation**

Exact Match is strict.

Two answers may convey the same information but differ in wording.

For example:

- Reference: "Five years."
- Prediction: "A period of five years."

The answers are semantically equivalent, but may not receive an exact match depending on normalization.

F1 helps account for partial textual overlap, but it does not fully measure whether the extracted information is legally or contextually correct.

**5. What conclusions are justified?**

| **Claim** | **Supported by the results?** |
|---|---|
| The model is highly effective at rejecting unanswerable questions in the reported evaluations. | Yes |
| The model extracts answerable contractual information reliably. | No |
| Overall F1 alone adequately represents performance. | No |
| The answerability-aware approach addresses an important weakness in contract QA. | Yes, with limitations |
| The framework has solved legal contract information extraction. | No |

| | |
|---|---|
| D1–D6 mapping has been evaluated separately from QA extraction. | Yes |

**6. Central research insight**

The most important lesson from these experiments is that **information extraction and answerability detection must not be treated as the same capability.**

A system can be excellent at recognising when it should remain silent while still struggling to extract correct information when evidence exists.

The authors' results demonstrate this imbalance clearly.

The framework's strength is conservative rejection of unsupported answers. Its major weakness is extracting answerable contractual provisions.

That is the correct interpretation of the experimental findings—not that the model has achieved uniformly high contract intelligence performance.

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**PROMPT-5: After studying the framework, I want to examine whether its underlying methodological principles could be relevant outside lease-contract intelligence. Could answerability-aware extraction be useful in other business processes involving multiple documents or records? What characteristics would make another domain suitable for this approach? What changes would be required when transferring the methodology? Please distinguish transferable principles from domain-specific implementation details.**

Yes. The underlying principles of ROU-TransQA-v2 could be relevant to other business processes involving multiple documents or records. The key transferable idea is **not the IFRS 16 framework itself, but the separation of answer extraction from answerability assessment.**

A business system should not be forced to produce an answer simply because a user asks a question. It should first establish whether the available records contain sufficient evidence to support that answer.

**1. Where could answerability-aware extraction be useful?**

Consider business processes where employees need to retrieve information from several documents, databases, or records.

**Procurement and vendor management**

Question: What payment terms were agreed with this supplier?

Evidence could come from vendor agreements, purchase orders and amendments.

If no document establishes the payment terms, the system should abstain rather than infer them from a previous transaction.

**Human resources**

Question: What is the employee's approved notice period?

Evidence could come from appointment letters, employment agreements and amendments.

The system must distinguish an approved contractual term from a general HR policy.

**Banking and compliance**

Question: Has the required approval been obtained?

Evidence could come from approval records, workflow logs and compliance documents.

An approval request being submitted is not evidence that approval was granted.

**IT service management**

Question: Has this incident been resolved?

Evidence could come from incident tickets, work notes, change records and closure logs.

A technician's note that a fix was attempted does not necessarily establish successful resolution.

**Finance and accounting**

Question: Has this invoice been paid?

Evidence could come from invoices, payment records, bank statements and reconciliation reports.

An invoice marked as processed does not necessarily prove that payment was completed.

These are potential applications of the methodology, not applications validated by the paper.

**2. What makes a domain suitable?**

Not every business process needs answerability-aware extraction. A domain is particularly suitable when it has the following characteristics.

| **Characteristic** | **Why it matters** |
|---|---|
| Multiple information sources | Relevant evidence is distributed across documents or records. |
| Structured questions | Users repeatedly ask identifiable questions about the available information. |
| Missing information | Some questions genuinely cannot be answered from the available records. |
| High cost of incorrect answers | Unsupported answers can cause financial, operational or compliance problems. |
| Traceability requirements | Users need to know which document or record supports an answer. |
| Defined answer criteria | Domain experts can establish what counts as sufficient evidence. |
| Repeated workflows | The same types of questions occur frequently enough to justify automation. |

The most important characteristic is the existence of **meaningful unanswerable cases**.

If every question can be answered directly from a complete and reliable database, a separate answerability mechanism may provide limited additional value.

Conversely, when information is incomplete, inconsistent or distributed across multiple sources, answerability detection becomes much more valuable.

**3. Which principles are transferable?**

The framework contains several methodological principles that can be reused independently of lease accounting.

**Transferable methodological principles**

1. **Separate answerability detection from answer extraction**

Determine whether sufficient evidence exists before attempting to generate an answer.

2. **Use evidence-grounded extraction**

Extract answers from actual source material rather than relying on plausible assumptions or general knowledge.

3. **Support abstention**

Allow the system to explicitly indicate that the available evidence does not support an answer.

4. **Preserve evidence traceability**

Retain the source document, relevant passage or record associated with each answer.

5. **Evaluate answerable and unanswerable cases separately**

Measure extraction quality and rejection quality independently so that one does not hide weaknesses in the other.

6. **Map extracted evidence to domain-specific indicators**

Convert validated evidence into business categories, classifications or analytical indicators.

These principles are methodological. They do not depend on IFRS 16 or lease agreements.

**4. Which implementation details are domain-specific?**

This is an important distinction. Transferring the framework does not mean copying its entire implementation.

| **Component** | **In the original paper** | **What changes in another domain?** |
|---|---|---|
| Input data | Lease-related legal contracts and CUAD benchmark | Domain-specific documents and records |
| Questions | Contractual questions | Questions relevant to the business process |
| QA model | RoBERTa-based extractive QA | Same model may be tested, adapted or replaced |
| Answerability | Explicit answerability classification | Domain-specific criteria for sufficient evidence |
| Training | Conditional masking of span supervision for unanswerable instances | Recreate and annotate training examples for the new domain |
| Indicator mapping | Expert-informed D1–D6 mapping | New business categories and classification rules |
| Analytics | IFRS 16-oriented multidimensional analytics | Domain-specific reporting and decision support |
| Evaluation | EM, F1, HasAns F1 and NoAns F1 | Appropriate metrics plus domain-specific correctness and reliability measures |

The original paper's CUAD dataset, D1–D6 indicators and IFRS 16 mapping are not transferable as-is.

RoBERTa is also not inherently mandatory. The important principle is the architecture's ability to distinguish supported answers from unsupported ones.

**5. What changes are required when transferring the methodology?**

A transfer would require more than replacing lease documents with business documents.

A practical adaptation would involve six stages.

1. **Define the new business problem**

Identify the questions employees need answered and the decisions that depend on those answers.

Example: In procurement, identify questions about payment terms, delivery obligations and approval status.

2. **Build a domain-specific dataset**

Collect representative documents and records. Include questions whose answers are present and questions whose answers are genuinely absent.

3. **Define answerability criteria**

Establish what evidence is sufficient, insufficient, contradictory or outdated.

This step requires domain experts because the meaning of sufficient evidence varies across business processes.

4. **Adapt and train the extraction model**

Fine-tune the QA system on the new questions and documents. Adapt answerability classification and span supervision to the new data.

5. **Create domain-specific indicator mapping**

Define the business categories to which extracted evidence should be mapped. For procurement, these might include payment terms, delivery obligations and approval status.

6. **Evaluate the adapted system**

Evaluate answerable and unanswerable cases separately. Also test source attribution, conflicting records, outdated information and performance across different document types.

**An additional challenge: multiple documents may disagree**

The original idea becomes more complex when information comes from several records.

For example:

- Original vendor agreement: payment due in 30 days.
- Later amendment: payment due in 45 days.
- Purchase order: payment due in 30 days.

The system cannot simply extract the first matching answer.

It must account for:

- Document dates and effective dates.
- Whether an amendment supersedes an earlier agreement.
- Document authority and reliability.
- Conflicting evidence.
- Whether the available records establish the current applicable term.

This introduces a requirement for **cross-document evidence reconciliation**.

Answerability alone is not enough. A system may find evidence but still lack sufficient grounds to determine which evidence is authoritative.

A practical extension would distinguish at least:

- Answerable: sufficient and consistent evidence exists.
- Unanswerable: required evidence is absent.
- Conflicting or unresolved: evidence exists, but the records do not establish a single reliable answer.

This is a proposed extension for multi-document business processes, not a claim that the original paper implements these three categories.

**6. Final distinction: what transfers and what does not?**

| **Transferable** | **Domain-specific** |
|---|---|
| Answerability-aware extraction | IFRS 16 accounting rules |
| Evidence-based answering | Lease-related questions |

| | |
|---|---|
| Abstention when evidence is insufficient | CUAD dataset |
| Conditional span supervision concept | Original training annotations |
| Evidence traceability | D1–D6 indicator definitions |
| Separate answerable/unanswerable evaluation | Lease-specific analytics |
| Expert-guided mapping | Accounting measurement rules |

**Central conclusion:** The framework's broader significance lies in its methodological approach to reliable information extraction. Its principles are relevant wherever business decisions depend on retrieving evidence from incomplete or distributed records. However, successful transfer requires new datasets, domain-specific answerability definitions, appropriate evidence-reconciliation rules, and new indicator mappings. The original experimental results do not establish performance in those other domains.

Consider the principles identified in the paper: answerability, evidence-based extraction and handling uncertainty

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**PROMPT-6: Consider the principles identified in the paper: answerability, evidence-based extraction and handling uncertainty. I want to understand whether these principles could be relevant to financial operations. In financial operations, information may need to be compared across internal records and external confirmations. What additional challenges arise in this setting compared with extracting information from a single lease contract? Explain the differences before proposing any adaptation**.

The principles of answerability, evidence-based extraction and uncertainty handling are relevant to financial operations. However, financial operations introduce a more complex problem than extracting information from a single lease contract.

The central difference is this:

**In a single-contract setting, the system primarily determines whether a document contains evidence answering a question. In financial operations, it must determine whether information from multiple independent sources agrees, is authoritative, and represents the same underlying transaction.**

**1. Difference between the two settings**

Consider the original paper's setting and a financial reconciliation process.

**Single lease-contract extraction**

**Lease agreement**

One primary evidence source

**Question**

Does the agreement specify the lease commencement date?

**Answerability assessment and extraction**

Is the information present and supported by the contract?

**Financial operations**

**Internal records**

Ledger, payment system, transaction logs

**External records**

Bank statements, confirmations, vendor statements

**Cross-source comparison**

Do the records refer to the same transaction, and do their details agree?

**Evidence assessment and reconciliation**

Is there sufficient evidence to confirm the transaction, explain discrepancies or leave it unresolved?

The second setting requires more than finding an answer in text. It requires comparing information across sources.

**2. Additional challenges in financial operations**

**A. Multiple sources may contain different versions of the same information**

Suppose a company records a payment of ₹50,000 in its internal system, while the bank statement shows ₹48,500.

Possible explanations include:

- Bank charges of ₹1,500.
- A partial settlement.
- An incorrect internal entry.
- A transaction that has not been fully processed.

A system extracting information from one document might successfully identify both amounts. But extraction alone cannot explain why they differ.

**Additional challenge:** The system must identify and investigate discrepancies, not merely extract values.

**B. Identifying whether two records refer to the same transaction**

Consider these records:

| Field | Internal record | Bank confirmation |
|---|---|---|
| Amount | ₹50,000 | ₹50,000 |
| Date | 10 October | 11 October |
| Reference | TXN-1045 | BANK-7782 |
| Description | Vendor payment | NEFT transfer |

The amount is identical, but the dates and reference numbers differ.

Are these records describing the same transaction?

The system must use multiple attributes to establish a reliable match.

This introduces **entity resolution and transaction matching**, which are not central to single-document extractive QA.

**C. Different sources have different levels of authority**

Not every source should be treated equally.

For example:

- An internal payment request establishes that payment was requested.
- An internal ledger entry establishes what the company recorded.
- A bank confirmation may establish that funds were transferred.
- A vendor confirmation may establish that the vendor received or acknowledged payment.

These sources answer different questions.

A payment request cannot establish that money was transferred. Similarly, an internal ledger entry alone may not independently confirm that the external transaction occurred.

**Additional challenge:** The system must understand what each source can legitimately establish.

**D. Timing differences**

Financial information changes over time.

For example:

- A payment is initiated on Monday.
- It is debited by the bank on Tuesday.
- The vendor receives it on Wednesday.
- The internal ledger is updated on Thursday.

All four records may be correct.

A system that treats them as contradictory without considering transaction lifecycle and timestamps could generate false exceptions.

**Additional challenge:** It must understand effective dates, processing dates, posting dates and settlement dates.

**E. Missing evidence is different from conflicting evidence**

This distinction is particularly important.

| **Situation** | **Interpretation** |
|---|---|
| No bank confirmation exists | Evidence is missing |
| Bank confirms ₹50,000; internal record shows ₹48,500 | Evidence conflicts |
| Bank confirmation is pending | Evidence is incomplete |
| Two bank statements show different transaction statuses | Evidence requires investigation |
| All relevant records agree | Evidence supports reconciliation |

A simple answerable/unanswerable classification cannot adequately represent all these situations.

A question can have available evidence without having a sufficiently reliable or consistent answer.

**F. Financial data is not always natural language**

The original framework uses extractive question answering on contractual text.

Financial operations also involve:

- Transaction IDs.
- Dates and timestamps.
- Currency values.
- Debit and credit entries.
- Account numbers.
- Status codes.
- Structured database fields.

- Tables and spreadsheets.

These require structured-data processing, numerical comparison and transaction matching in addition to NLP.

**G. The consequences of incorrect conclusions are different**

In financial operations, an incorrect conclusion may result in:

- Incorrect reconciliation.
- Duplicate payments.
- Unidentified outstanding balances.
- Incorrect financial reporting.
- Audit exceptions.
- Compliance issues.

Therefore, the system must provide not just an answer, but a defensible explanation supported by evidence.

**3. Comparing the methodological requirements**

| **Dimension** | **Single lease contract** | **Financial operations** |
|---|---|---|
| Evidence source | Primarily one contract | Multiple internal and external sources |
| Main task | Find contractual evidence | Extract, match and reconcile evidence |
| Answerability | Is the answer present? | Is there sufficient evidence across sources? |
| Uncertainty | Missing or insufficient contractual evidence | Missing, conflicting, delayed or unreliable records |
| Authority | Interpret contractual clauses | Determine source authority and evidentiary role |
| Temporal reasoning | Contractual dates and conditions | Transaction lifecycle and changing statuses |
| Data format | Primarily legal text | Text, tables, databases and transaction records |
| Output | Answer with supporting clause | Reconciled result, discrepancy or unresolved case |

**4. What this means for the original principles**

The three principles remain relevant, but their interpretation becomes broader.

**Answerability**

In contract extraction, the question is whether the document contains the requested information.

In financial operations, the question becomes whether the available records collectively establish the requested financial fact.

**Evidence-based extraction**

In contract extraction, the system identifies a supporting passage.

In financial operations, it must identify the relevant fields and records, establish their relationship, and preserve the evidence supporting the conclusion.

**Handling uncertainty**

In contract extraction, uncertainty often concerns whether an answer can be extracted.

In financial operations, uncertainty can arise even when all the required data is present. The data may conflict, be outdated, refer to different transactions or represent different stages of processing.

**5. The fundamental conceptual difference**

There are three distinct questions a financial operations system must answer:

1. **Extraction:** What information does each source contain?
2. **Matching and reconciliation:** Do the records refer to the same transaction, and do their details agree?
3. **Evidence sufficiency:** Is there enough reliable evidence to establish the financial conclusion?

These questions should not be collapsed into one answerability decision.

For example, a system may extract ₹50,000 from an internal ledger and ₹48,500 from a bank statement perfectly. It has succeeded at extraction, but reconciliation remains unresolved.

Likewise, matching two records does not prove that the underlying transaction was completed.

**The key insight:** Moving from lease-contract intelligence to financial operations changes the problem from primarily *document-level evidence extraction* to *multi-source evidence evaluation and reconciliation*. Answerability remains useful, but it must account for source authority, transaction identity, time, numerical consistency and conflicting evidence.

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**PROMPT-7: I want to examine a specific scenario. An internal financial trade record contains a settlement date, but the corresponding external confirmation does not contain that information. Based on the answerability principle in the research paper, how should an automated system handle this situation? Should the system classify it as agreement, disagreement or insufficient evidence? Explain the risks of treating missing information as confirmation.**

The correct classification is **insufficient evidence**, not agreement.

The internal trade record contains a settlement date, but the external confirmation does not. Therefore, the two sources cannot be said to agree or disagree about that date.

**1. Apply the answerability principle**

Suppose the records contain:

| Field | Internal trade record | External confirmation |
|---|---|---|
| Trade reference | TR-1025 | TR-1025 |
| Trade amount | ₹5,00,000 | ₹5,00,000 |
| Settlement date | 10 November | Not provided |

Now consider the question:

**Does the external confirmation support the settlement date recorded internally?**

The system should reason as follows:

1. The internal record explicitly provides 10 November.
2. The external confirmation contains no settlement date.
3. There is no external evidence confirming 10 November.
4. There is also no external evidence contradicting 10 November.

Therefore, the result is insufficient evidence.

**2. Why not agreement or disagreement?**

**Agreement**

Not applicable

Both sources provide compatible settlement dates.

Example: Internal = 10 November; External = 10 November.

**Disagreement**

Not applicable

Both sources provide dates, but the dates differ.

Example: Internal = 10 November; External = 12 November.

**Insufficient evidence**

Correct classification

One source provides a date, while the other does not provide enough information to compare.

Example: Internal = 10 November; External = Not provided.

The absence of a value is not a contradictory value.

**3. Risks of treating missing information as confirmation**

If the automated system treats the missing external settlement date as agreement, several problems arise.

**1. False confirmation**

The system could report that the external counterparty confirmed 10 November, even though the confirmation never stated that date.

**2. Incorrect reconciliation**

The trade could be marked as successfully reconciled when an important field remains unverified.

**3. Loss of audit traceability**

An auditor examining the external confirmation would find no evidence supporting the system's conclusion.

**4. Concealed operational risk**

A settlement date mismatch might remain undetected until settlement fails or a financial discrepancy occurs.

**5. Unjustified confidence**

The system would convert an unknown value into a confirmed value, making its output appear more reliable than the underlying evidence.

**4. How should the automated system represent the result?**

A suitable output would be:

**Settlement date verification**

Insufficient evidence

| **Attribute** | **Result** |
|---|---|
| Internal settlement date | 10 November |
| External settlement date | Not provided |
| Comparison result | Cannot be established |

| | |
|---|---|
| Reconciliation status | Pending verification |

**System explanation**

"The internal trade record specifies 10 November as the settlement date. The external confirmation does not contain a settlement date. Therefore, external confirmation of the settlement date cannot be established."

The system should preserve the internal date as recorded, but must not label it externally confirmed. The missing field should be flagged for verification rather than automatically treated as an exception or a successful match.

**5. Connection to the research paper**

This scenario illustrates the paper's central answerability principle:

- Extract only information supported by the available evidence.
- Do not generate an answer when the evidence does not support one.
- Distinguish absence of evidence from evidence that contradicts a claim.

There is one important distinction: the paper's answerability classification concerns whether a question can be answered from the supplied contractual evidence. Applying it to financial reconciliation requires an additional comparison layer that distinguishes agreement, disagreement and insufficient evidence.

**Final conclusion:** Missing external information must remain unknown. It cannot be converted into agreement simply because the internal record contains a value.

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**PROMPT-8: I want to investigate post-trade reconciliation as a possible operational context for these ideas. Explain how internal OMS or portfolio accounting records are compared with external broker or custodian records. What types of discrepancies can occur? How is identifying a matching trade different from extracting an answer from a contract? Could answerability-aware evidence handling help when investigating incomplete or ambiguous trade records?**

**1. What is post-trade reconciliation?**

Post-trade reconciliation is the process of comparing an organisation's internal records of executed trades with records received from external parties, such as brokers and custodians.

The objective is to establish whether both sides have recorded the same transaction and whether its important details are consistent.

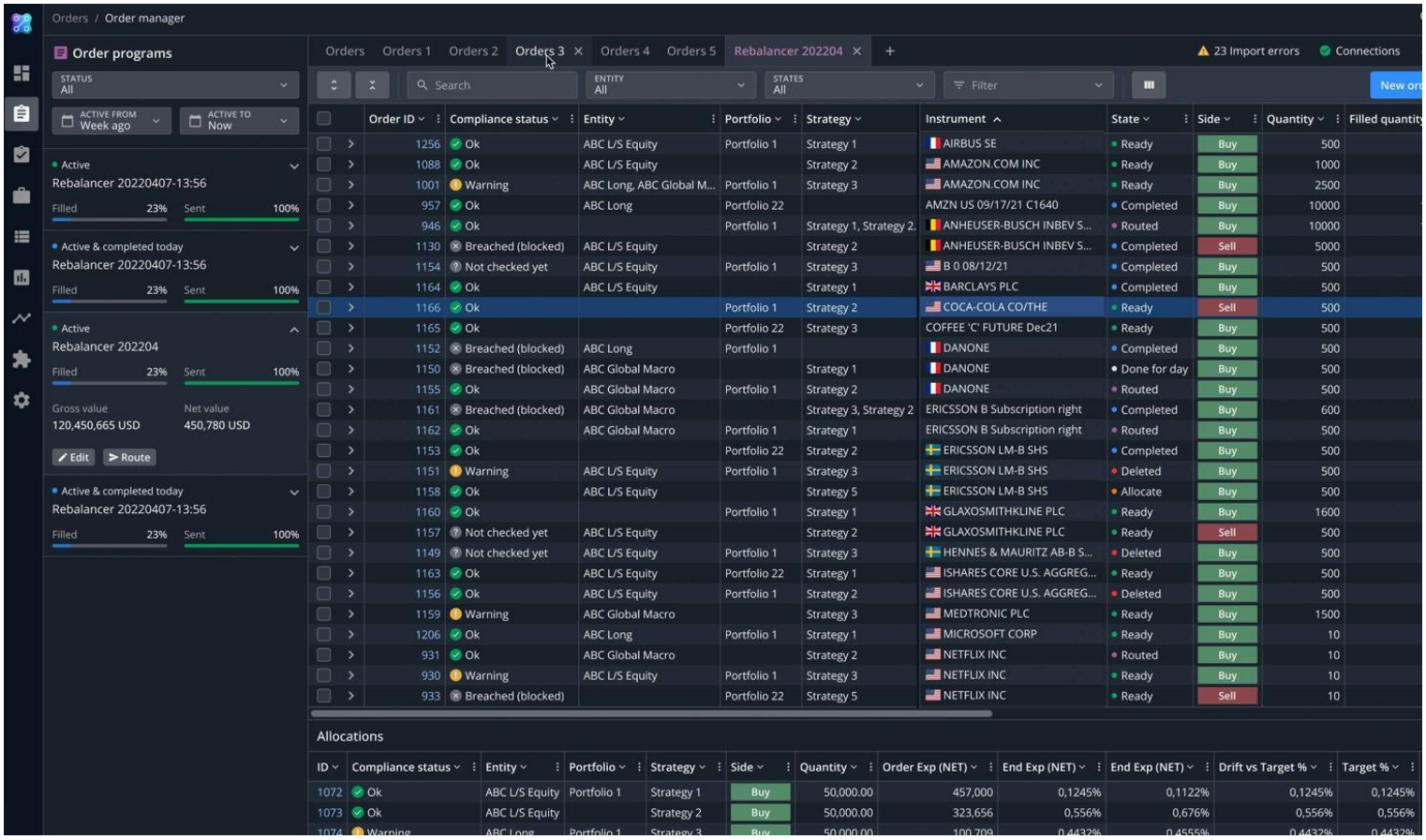
For example, an investment firm executes a purchase of shares through a broker. The firm's internal system records the trade. The broker sends a trade confirmation, and the custodian maintains records associated with settlement and asset holdings.

These records should correspond, but differences can arise because the systems capture different stages of the transaction.

**2. Which systems and records are compared?**

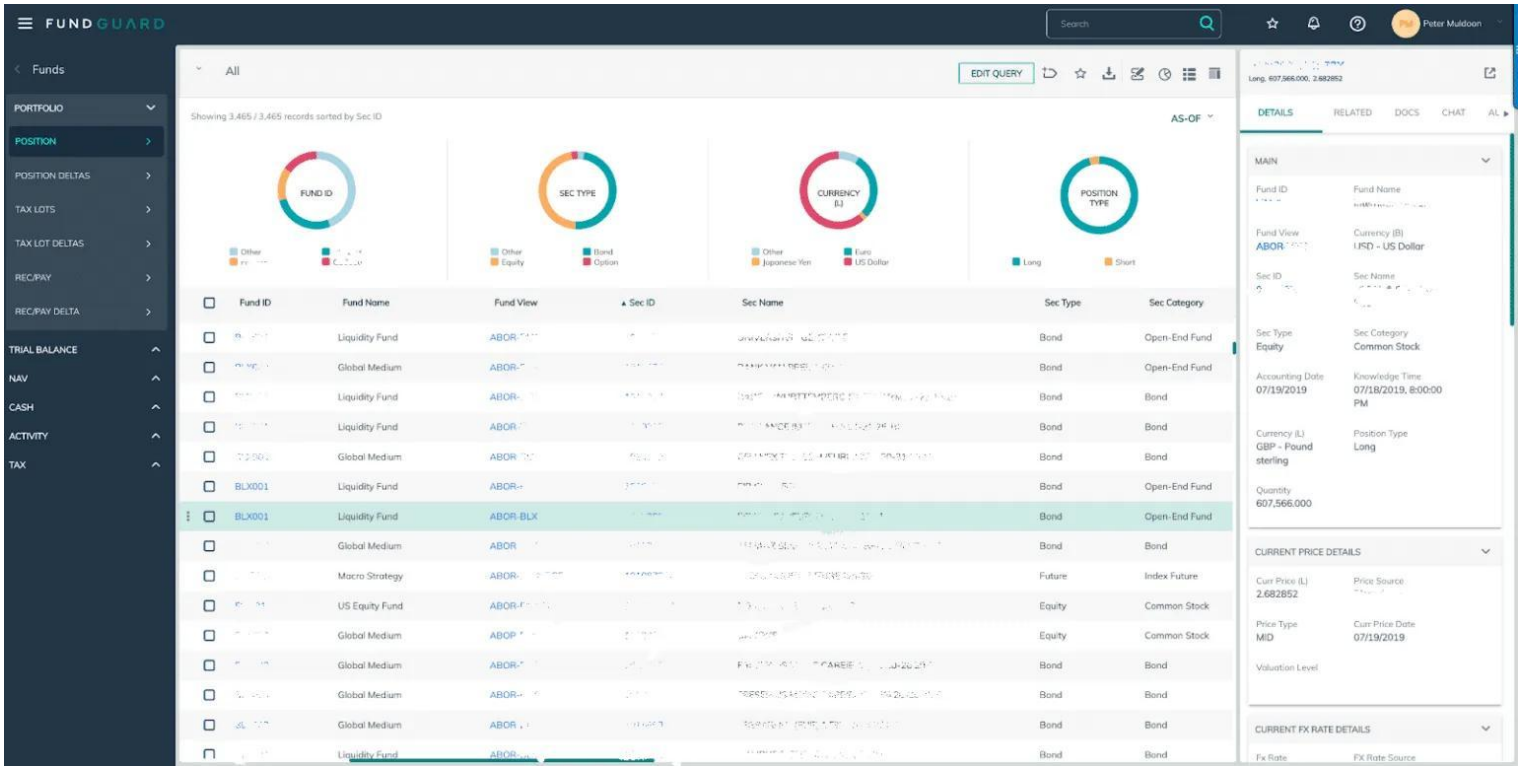
There are three important participants.

**A. Internal systems**




**Order Management System (OMS)**

Records orders and trade execution details, such as security, quantity, price, broker and execution time.




**Portfolio Accounting System**

Maintains accounting records associated with investments, including positions, cash balances, transaction values and settlement-related entries.

**B. External records**

InteractiveBrokers

Trade Confirmation Report
May 3, 2022 - May 18, 202

Interactive Brokers LLC, Two Pickwick Plaza, Greenwich, CT 06830

Expand All | Contract All

> Account Information

v Trades

| Acct ID | Symbol | Trade Date/Time | Settle Date | Exchange | Type | Quantity | Price | Proceeds | Comm | Fee | Code |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Stocks** | | | | | | | | | | | |
| **USD** | | | | | | | | | | | |
| UXXXXXX | VPU | 2017-07-05, 12:05:22 | 2017-07-10 | - | SELL | -300 | 112.9607333 | 33,888.22 | -2.02 | 0.00 | P |
| **Total VPU (Sold)** | | | | | | **-300** | **112.9607333** | **33,888.22** | **-2.02** | **0.00** | |
| UXXXXXX | XLF | 2017-07-05, 12:04:57 | 2017-07-10 | - | SELL | -2,000 | 25.031575 | 50,063.15 | -10.69 | 0.00 | P |
| **Total XLF (Sold)** | | | | | | **-2,000** | **25.031575** | **50,063.15** | **-10.69** | **0.00** | |
| UXXXXXX | XLU | 2017-07-05, 12:05:07 | 2017-07-10 | - | SELL | -1,000 | 51.38052 | 51,380.52 | -5.31 | 0.00 | P |
| **Total XLU (Sold)** | | | | | | **-1,000** | **51.38052** | **51,380.52** | **-5.31** | **0.00** | |
| **Total** | | | | | | | | **135,331.89** | **-18.02** | **0.00** | |
| **Acct ID** | **Symbol** | **Trade Date/Time** | **Settle Date** | **Exchange** | **Type** | **Quantity** | **Price** | **Notional Value** | **Comm** | **Fee** | **Code** |
| **Futures** | | | | | | | | | | | |
| **USD** | | | | | | | | | | | |
| UXXXXXX | ESU7 | 2017-07-06, 15:58:38 | 2017-07-07 | - | BUY | 2 | 2,408.0000 | -240,800.00 | -4.08 | 0.00 | C |
| UXXXXXX | ESU7 | 2017-07-10, 09:20:41 | 2017-07-11 | - | BUY | 1 | 2,420.7500 | -121,037.50 | -2.04 | 0.00 | C |
| UXXXXXX | ESU7 | 2017-07-10, 09:20:53 | 2017-07-11 | - | BUY | 2 | 2,420.5000 | -242,050.00 | -4.08 | 0.00 | C |
| UXXXXXX | ESU7 | 2017-07-11, 11:46:08 | 2017-07-12 | - | BUY | 2 | 2,418.0000 | -241,800.00 | -4.08 | 0.00 | C |
| UXXXXXX | ESU7 | 2017-07-13, 18:15:03 | 2017-07-17 | - | BUY | 4 | 2,445.5000 | -489,100.00 | -8.16 | 0.00 | C |
| UXXXXXX | ESU7 | 2017-07-13, 18:15:31 | 2017-07-17 | - | BUY | 4 | 2,445.5000 | -489,100.00 | -8.16 | 0.00 | C;P |
| **Total ESU7 (Bought)** | | | | | | **15** | **2,431.8500** | **-1,823,887.50** | **-30.60** | **0.00** | |
| UXXXXXX | ESU7 | 2017-07-05, 13:20:29 | 2017-07-06 | - | SELL | -2 | 2,429.7500 | 242,975.00 | -4.08 | 0.00 | O |
| UXXXXXX | ESU7 | 2017-07-05, 13:22:23 | 2017-07-06 | - | SELL | -2 | 2,430.2500 | 243,025.00 | -4.08 | 0.00 | O |
| UXXXXXX | ESU7 | 2017-07-05, 13:35:00 | 2017-07-06 | - | SELL | -4 | 2,430.5000 | 486,100.00 | -8.16 | 0.00 | O |
| UXXXXXX | ESU7 | 2017-07-06, 10:37:39 | 2017-07-07 | - | SELL | -1 | 2,412.2500 | 120,612.50 | -2.04 | 0.00 | O |
| UXXXXXX | ESU7 | 2017-07-06, 12:32:46 | 2017-07-07 | - | SELL | -1 | 2,416.5000 | 120,825.00 | -2.04 | 0.00 | O |
| UXXXXXX | ESU7 | 2017-07-06, 15:46:27 | 2017-07-07 | - | SELL | -5 | 2,407.0000 | 601,750.00 | -10.20 | 0.00 | O;P |
| UXXXXXX | ESU7 | 2017-07-14, 10:39:13 | 2017-07-17 | - | SELL | -5 | 2,449.5000 | 612,375.00 | -10.20 | 0.00 | O |
| **Total ESU7 (Sold)** | | | | | | **-20** | **2,427.6625** | **2,427,662.50** | **-40.80** | **0.00** | |
| **Total ESU7** | | | | | | **-5** | | **603,775.00** | **-71.40** | **0.00** | |
| **Total** | | | | | | | | **603,775.00** | **-71.40** | **0.00** | |

> Financial Instrument Information

> Codes

v Notes/Legal Notes

**Broker confirmation**

Provides details of the executed trade, such as execution price, quantity, trade date, commission and transaction reference.

# NCSS REPORT:

National Clearing Company of Pakistan Limited.
Change Settling Information Report

CM ID: 04234
Date Range: From 11/01/2016 To 11/01/2016
Settling Bank: ALL
Branch: ALL

Page: 1 of 1
User Id: C0423401
Date: 11/01/2016
Time: 05:24:22

| CM ID | CM Name | Main SB Branch | Main SB A/c | Changed SB | Changed SB Branch | Changed SB A/c | Change Date | For Sett. Date |
|---|---|---|---|---|---|---|---|---|
| 04234 | KSE Corporate BCM (Regular) 04234 | ABK/KSE | 04234 | BAF | KSE | 04234-02 | 08/01/2016 | 11/01/2016 |

End of Report

**Custodian records**

Provide information about securities holdings, cash movements, settlement status and asset positions maintained by the custodian.

An important distinction: a broker confirmation and a custodian record do not necessarily contain identical information. The broker primarily confirms execution details, while the custodian records custody and settlement-related information.

**3. How does reconciliation work?**

Consider an investment firm purchasing 100 shares of a company.

Its internal OMS records the executed trade. The firm subsequently receives an external broker confirmation.

The reconciliation process typically follows these stages:

1. **Trade execution**

The order is executed through the broker.

2. **Internal recording**

The OMS records the execution. Portfolio accounting may also receive the transaction.

3. **External confirmation**

The broker sends a confirmation containing the execution details.

4. **Data normalisation**

Records are standardised into comparable fields, including dates, currencies, security identifiers and quantities.

5. **Trade matching**

The system identifies which internal and external records represent the same trade.

6. **Field-level comparison**

It compares the relevant details and identifies missing values, differences or inconsistencies.

7. **Exception investigation**

Unmatched or inconsistent records are flagged for investigation and resolution.

**Example of a matching trade**

| **Field** | **Internal OMS** | **Broker confirmation** |
|---|---|---|
| Security | ABC Ltd | ABC Ltd |
| Trade date | 3 Nov | 3 Nov |
| Quantity | 100 | 100 |
| Price | ₹500 | ₹500 |
| Currency | INR | INR |
| Settlement date | 5 Nov | Not provided |

The trade appears to match on the available identifying and execution fields.

However, the settlement date cannot be externally verified from this confirmation.

This distinction becomes important when designing an automated reconciliation system.

**4. What types of discrepancies can occur?**

Discrepancies are not all the same. They can arise from missing information, incorrect values, timing differences or problems identifying the underlying transaction.

| **Discrepancy type** | **Example** | **What it means** |
|---|---|---|
| Missing field | External record has no settlement date | Information cannot be compared |
| Quantity mismatch | Internal: 100; external: 90 | Trade details differ |
| Price mismatch | Internal: ₹500; external: ₹502 | Execution details differ |
| Date mismatch | Trade date differs | Could be an error or a date-convention issue |
| Currency mismatch | Internal: USD; external: EUR | Requires investigation |
| Unmatched trade | Internal trade has no corresponding external record | External record may be delayed or absent |

| | | |
|---|---|---|
| Duplicate trade | One internal trade matches two external records | Could indicate duplicate reporting or multiple executions |
| Status mismatch | Internal: settled; custodian: pending | Systems report different lifecycle states |
| Reference mismatch | Different transaction IDs | Matching requires other identifying attributes |

There are also more subtle cases.

**A. Timing differences**

An internal system might record a trade immediately after execution, while an external confirmation arrives later.

The absence of a confirmation at that moment does not prove that the trade was never executed.

**B. Partial executions**

Suppose an order is placed for 100 shares.

- Internal OMS: one order for 100 shares.
- Broker: two executions, one for 60 shares and another for 40 shares.

A simple one-to-one matching approach might incorrectly classify this as a mismatch.

The system needs to recognise that multiple execution records may collectively correspond to one internal order.

**C. Different transaction representations**

One system may represent a trade as a single transaction, while another reports separate execution legs, fees or settlement movements.

The records can be economically consistent without being textually identical.

**D. Lifecycle differences**

A trade can move through stages such as:

- Executed.
- Confirmed.
- Pending settlement.
- Settled.
- Failed or cancelled.

Two systems reporting different statuses do not necessarily mean one contains an incorrect record. They may have been updated at different times or use different status definitions.

**5. How is trade matching different from extracting an answer from a contract?**

This is the most important methodological distinction.

The two tasks may both involve identifying information from documents or records, but their underlying objectives differ.

**Contract information extraction**

Contract text:

"The lease commencement date is 1 April 2026."

Question: What is the commencement date?

**Extraction result**

1 April 2026, supported by the relevant contractual passage.

The primary objective is to identify a passage that answers a question.

**Post-trade matching**

**Internal record**

100 shares

₹500

3 November

**External record**

100 shares

₹502

3 November

**Matching result**

Same underlying trade is likely, but the execution price differs and requires investigation.

The primary objective is to identify corresponding records and assess whether their attributes are consistent.

The differences can be summarised as follows:

| **Dimension** | **Contract extraction** | **Trade reconciliation** |
|---|---|---|
| Input | Contractual text | Internal and external records |
| Main question | What does the document say? | Do these records describe the same trade? |
| Core operation | Question answering and span extraction | Record matching and field comparison |
| Evidence | Supporting text passage | Supporting records and field values |
| Missing information | Answer may be unanswerable | A field may be unverified |
| Conflicting information | Passage may not support the requested answer | Records may refer to the same trade but contain different values |
| Output | Extracted answer or abstention | Match, mismatch, unmatched or unresolved status |

**Trade matching is not the same as field agreement.**

Two records may match as the same transaction even though their prices or dates differ.

Likewise, two records may have identical amounts but refer to entirely different transactions.

Therefore, matching must come before meaningful field-level reconciliation.

**6. Could answerability-aware evidence handling help?**

Yes. It could be particularly useful during the investigation of incomplete or ambiguous trade records.

However, it should complement transaction matching and reconciliation rather than replace them.

The original paper's central principles can be adapted conceptually.

**A. Evidence-based extraction**

The system should extract only values actually present in the records.

For example:

- Internal settlement date: 5 November.
- External settlement date: Not provided.

It should preserve the internal date without claiming that the external source confirms it.

**B. Answerability-aware handling**

The system should distinguish between:

- Information explicitly available.
- Information absent from the source.
- Information that exists but is ambiguous.
- Information that conflicts with another source.

For financial reconciliation, this requires a richer outcome model than a simple answerable/unanswerable classifier.

**C. Evidence traceability**

Every conclusion should be linked to its source.

For example:

| **Conclusion** | **Supporting evidence** |
|---|---|
| Trade quantity is 100 | Internal OMS record and broker confirmation |
| Execution price is ₹500 internally | OMS record |
| Execution price is ₹502 externally | Broker confirmation |
| Settlement date is 5 November internally | OMS record |
| External settlement date is unverified | No date in broker confirmation |

This makes the investigation reproducible and auditable.

**7. A possible evidence classification for post-trade operations**

A useful conceptual extension would distinguish five outcomes:

**1. Matched and consistent**

Verified match

Both records refer to the same trade and the compared fields agree.

**2. Matched with discrepancy**

Exception

The records refer to the same trade, but one or more values differ.

**3. Incomplete evidence**

Unverified

A required field is missing from one or more sources.

**4. Ambiguous or unresolved**

Investigation

Available evidence does not establish a reliable match or conclusion.

**5. Unmatched**

No corresponding record

No suitable corresponding record has been identified in the comparison set.

These are proposed operational classifications, not categories defined or experimentally validated by the ROU-TransQA-v2 paper.

Also, "unmatched" should not automatically mean that a trade does not exist. The external record could be delayed, missing or represented differently.

**8. The larger research opportunity**

The interesting research direction is to combine three capabilities:

**1. Information extraction**

Identify trade attributes from records.

**2. Transaction matching**

Determine which internal and external records correspond to the same trade.

**3. Evidence sufficiency and uncertainty assessment**

Determine whether the available records support a reliable reconciliation conclusion.

**Evidence-grounded reconciliation**

Verified matches, discrepancies and unresolved exceptions, each with traceable evidence.

The three capabilities solve different problems:

- Extraction tells us what each record says.
- Matching tells us which records belong together.
- Evidence assessment tells us what conclusions those records actually justify.

**The central insight:** Answerability-aware evidence handling could make post-trade reconciliation more reliable by preventing missing, ambiguous or conflicting information from being converted into false confirmation. But the financial operations problem requires additional capabilities—especially transaction matching, temporal reasoning, numerical comparison and source-authority assessment—that go beyond the original contract-QA framework.

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**PROMPT-9: Based on the supplied research paper and the post-trade reconciliation workflow, examine which methodological principles can be adapted to a system called ReconIQ. Consider: - Answerability-aware evidence handling. - Missing information. - Ambiguous evidence. - Evidence-supported extraction. - Separation of extraction and validation. For each principle, explain: 1. What the original paper does. 2. How the principle could apply to ReconIQ. 3. What modifications are necessary. 4. What cannot be transferred directly. Do not claim that the paper studies trade reconciliation.**

**Adapting ROU-TransQA-v2 principles to ReconIQ**

The original paper, *An Answerability-Aware Contract Intelligence Framework for Right-of-Use Asset Management With IFRS 16-Oriented Indicator Mapping and Multidimensional Analytics*, studies contractual evidence extraction and IFRS 16-oriented indicator mapping.

ReconIQ, in the proposed post-trade reconciliation context, has a different operational objective: compare internal trade records with external broker or custodian records, identify discrepancies and support their investigation.

The paper does not study trade reconciliation. Therefore, the appropriate approach is to transfer its **methodological principles**, not claim that its original framework can be directly deployed in ReconIQ.

The distinction is important:

- The paper asks whether contractual evidence supports an answer to a question.
- ReconIQ needs to establish whether records correspond to the same trade and whether their information supports a reconciliation conclusion.

The following analysis separates what the paper actually implements from what would need to be designed for ReconIQ.

**1. Answerability-aware evidence handling**

**1. What the original paper does**

ROU-TransQA-v2 combines RoBERTa-based extractive question answering with an explicit answerability classification component.

It distinguishes questions for which the supplied contract contains an answer from questions for which it does not. For unanswerable instances, it uses conditional masking of span supervision during training so that the model is not trained to extract a nonexistent answer.

The objective is to prevent the system from producing unsupported contractual answers.

**2. How this could apply to ReconIQ**

ReconIQ could use the same principle when comparing internal and external trade records.

For example:

| **Field** | **Internal OMS** | **External broker** |
|---|---|---|
| Trade ID | TR-1025 | TR-1025 |
| Quantity | 100 | 100 |
| Price | ₹500 | ₹500 |
| Settlement date | 5 Nov | Missing |

ReconIQ should not report that the settlement dates agree. It should identify that the external evidence is insufficient to verify the internal date.

Answerability-aware handling would help ReconIQ distinguish between:

- A field supported by both sources.
- A field supported by only one source.
- A field unsupported by either source.
- A field for which available evidence conflicts.

**3. What modifications are necessary**

The original binary answerability decision would need to be extended into a field-level evidence assessment.

ReconIQ would need to assess:

- Whether the relevant field is present.
- Which source provides it.

- Whether the source is appropriate for verifying that field.
- Whether the values agree.
- Whether the evidence is sufficiently reliable to support a conclusion.

It should also distinguish a missing value from a failed transaction match.

**4. What cannot be transferred directly**

The paper's contract-question answerability classifier cannot simply be reused as a trade-reconciliation classifier.

Its answerable/unanswerable labels are designed around contractual questions and CUAD annotations. ReconIQ needs labels and decision rules designed around trade records, matching outcomes and reconciliation exceptions.

**2. Missing information**

**1. What the original paper does**

The paper explicitly handles unanswerable questions. When the requested information is absent from the contract, the system should reject the request rather than extract an unsupported answer.

Its training approach conditionally masks span supervision for unanswerable examples.

This is an important principle: the absence of an answer must be represented explicitly, rather than forcing the extraction model to select an irrelevant passage.

**2. How this could apply to ReconIQ**

Missing information is common in reconciliation records.

For example:

- Internal record contains a settlement date.
- Broker confirmation contains no settlement date.
- Custodian record has not yet arrived.

ReconIQ should preserve the available internal value while recording that external verification is unavailable.

It should not:

- Copy the internal value into the external field.
- Assume the external party agrees.
- Mark the field as successfully reconciled.
- Automatically classify the missing value as a discrepancy.

**3. What modifications are necessary**

ReconIQ would need to distinguish different reasons for missing information:

| **Missing-information condition** | **Suggested interpretation** |
|---|---|
| Field absent from the document | Not provided |
| External confirmation not received | Awaiting external evidence |
| Field not applicable to the transaction | Not applicable |
| Record received but field is blank | Incomplete record |

| | |
|---|---|
| Required source unavailable | Verification unavailable |

These distinctions matter because they require different operational responses.

For example, a missing settlement date in a broker confirmation may be acceptable if that document type is not expected to provide it. The same omission in a source expected to confirm settlement details may require investigation.

**4. What cannot be transferred directly**

The paper's treatment of unanswerable QA examples does not provide a complete model for financial data missingness.

In ReconIQ, missingness depends on field requirements, source type, transaction lifecycle and timing. It may be temporary, legitimate or operationally significant.

Therefore, the paper's no-answer mechanism is a conceptual starting point, not a complete missing-data management solution.

**3. Ambiguous evidence**

**1. What the original paper does**

The paper focuses on answerable and unanswerable contractual questions. Its central design is explicit answerability classification combined with extractive QA.

An important interpretive distinction is that related text does not necessarily establish the proposition asked in a question.

For example, a contract mentioning a possible lease extension does not necessarily establish an unconditional extension right.

However, the paper's abstract does not establish that ambiguous evidence is implemented as an independent third classifier category. It should not be represented as one of the paper's confirmed formal labels.

**2. How this could apply to ReconIQ**

ReconIQ can encounter evidence that exists but does not establish a clear reconciliation conclusion.

Example:

| **Source** | **Settlement status** |
|---|---|
| Internal portfolio accounting | Settled |
| Custodian record | Processed |
| External confirmation | No settlement status |

The word "processed" may not establish that settlement was completed.

ReconIQ should not automatically interpret these statuses as equivalent.

Another example is a partial execution:

- Internal OMS: 100 shares purchased.
- Broker records: 60 shares and 40 shares in separate executions.

The evidence is not necessarily contradictory. It may represent two executions belonging to one order.

**3. What modifications are necessary**

ReconIQ needs an explicit way to represent unresolved evidence.

It should consider:

- Whether two status terms have equivalent meanings.

- Whether records refer to the same transaction or different execution legs.
- Whether a field is incomplete or only partially supported.
- Whether multiple records provide conflicting information.
- Whether the evidence is sufficient to establish the requested conclusion.

A practical classification could distinguish:

1. Clear and consistent evidence.
2. Missing evidence.
3. Conflicting evidence.
4. Ambiguous or unresolved evidence.

The system should retain the reason for uncertainty, rather than simply returning a low-confidence answer.

**4. What cannot be transferred directly**

The original paper's answerability labels and contractual question annotations do not provide the semantic rules needed to interpret financial transaction statuses, partial executions or lifecycle events.

ReconIQ would need financial-domain definitions, status mappings and rules for resolving evidence across different systems.

**4. Evidence-supported extraction**

**1. What the original paper does**

The paper uses RoBERTa-based extractive question answering to identify answer spans in contractual text.

Its approach is evidence-oriented: the answer should be grounded in the supplied contract rather than generated from unsupported assumptions.

The framework also connects extracted contractual evidence to expert-informed D1–D6 indicators.

This creates a link between the original source evidence and the resulting accounting-oriented classification.

**2. How this could apply to ReconIQ**

ReconIQ should extract trade attributes directly from their respective sources and preserve the origin of each value.

For example:

| **Attribute** | **Extracted value** | **Evidence source** |
|---|---|---|
| Trade quantity | 100 | Internal OMS |
| Trade quantity | 100 | Broker confirmation |
| Execution price | ₹500 | Internal OMS |
| Execution price | ₹502 | Broker confirmation |
| Settlement date | 5 Nov | Internal OMS |
| Settlement date | Not provided | Broker confirmation |

This allows ReconIQ to explain exactly which records support each conclusion.

Instead of merely reporting "price mismatch", it can show:

- Internal price: ₹500.
- External price: ₹502.
- Internal source: OMS trade record.
- External source: Broker confirmation.

This is particularly valuable for auditability and exception investigation.

**3. What modifications are necessary**

ReconIQ would need to support more than textual answer extraction.

Its extraction layer should handle:

- Structured database fields.
- CSV and spreadsheet records.
- Broker confirmation documents.
- Custodian reports.
- Transaction identifiers.
- Numeric and date fields.
- Multiple executions associated with one order.

It should also preserve source metadata, such as record ID, document reference, timestamp and field location.

For numerical values, it may need normalisation before comparison. For example, different date formats, currency representations and security identifiers should be standardised without changing the original source values.

**4. What cannot be transferred directly**

RoBERTa-based extractive QA is designed to identify answers in text. It does not independently solve structured transaction extraction, entity resolution, numerical validation or cross-system data normalisation.

The paper's D1–D6 indicator mapping is also specific to its IFRS 16 application.

ReconIQ would need its own financial data schema and reconciliation attributes.

**5. Separation of extraction and validation**

**1. What the original paper does**

The paper combines extractive QA, answerability classification and a subsequent expert-informed indicator-mapping layer.

These components serve different purposes:

- Extractive QA identifies contractual answers.
- Answerability classification determines whether an answer is supported.
- D1–D6 mapping converts relevant evidence into accounting-oriented indicators.

The paper's architecture therefore provides a useful methodological example of separating evidence extraction from downstream interpretation.

However, its abstract does not establish a dedicated financial-style validation or transaction-reconciliation engine.

**2. How this could apply to ReconIQ**

This principle is particularly important for ReconIQ.

Extraction and validation should be separate stages.

Suppose the internal OMS reports a quantity of 100 shares, while the broker confirmation reports 90 shares.

The extraction layer should faithfully extract both values.

It should not decide that one is correct.

The validation layer should then compare the values and identify the discrepancy.

**Proposed ReconIQ processing structure**

**Internal and external records**

OMS, portfolio accounting, broker and custodian

**Stage 1: Evidence extraction**

Extract values and preserve their source references

**Stage 2: Answerability and evidence assessment**

Identify missing, ambiguous and unsupported information

**Stage 3: Trade matching**

Identify corresponding transactions across systems

**Stage 4: Validation and reconciliation**

Compare fields and identify discrepancies

**Stage 5: Exception reporting**

Verified matches, discrepancies and unresolved cases with supporting evidence

**3. What modifications are necessary**

ReconIQ should enforce a clear separation of responsibilities:

| **Component** | **Responsibility** |
|---|---|
| Extraction | Identify what each source actually reports |
| Evidence assessment | Determine whether information is present and sufficiently supported |
| Trade matching | Establish which records correspond to the same transaction |
| Validation | Compare values and apply reconciliation rules |
| Exception management | Record unresolved cases and supporting explanations |

This separation prevents the extraction model from silently correcting, inferring or modifying source information.

It also makes errors easier to investigate. If the system reports an incorrect reconciliation result, the organisation can determine whether the problem occurred during extraction, matching or validation.

**4. What cannot be transferred directly**

The paper's D1–D6 mapping layer cannot serve as ReconIQ's validation engine.

Accounting indicator classification and trade reconciliation are different tasks.

ReconIQ would require independently defined matching algorithms, financial validation rules, tolerance thresholds, transaction lifecycle logic and exception-management procedures.

**6. Consolidated comparison**

The following table summarises the five principles.

| Principle | Original paper | ReconIQ adaptation | Main modification |
|---|---|---|---|
| Answerability-aware evidence handling | Classifies contractual questions as answerable or unanswerable | Determines whether records support a reconciliation conclusion | Field-level and multi-source evidence assessment |
| Missing information | Rejects unanswerable questions and masks span supervision | Identifies absent, delayed and inapplicable trade information | Financial missing-data categories |
| Ambiguous evidence | Uses answerability-aware contractual QA; no separate ambiguity classifier established in the abstract | Identifies unclear statuses, partial executions and unresolved records | Financial semantic and lifecycle rules |
| Evidence-supported extraction | Extracts answers from contractual passages | Extracts trade attributes from internal and external records | Structured-data extraction and source traceability |
| Separation of extraction and validation | Combines QA, answerability and indicator mapping | Separates extraction, matching, comparison and reconciliation | Dedicated matching and validation components |

**7. What ReconIQ can take from the paper**

**Transferable methodological foundation**

- Never produce an answer unsupported by available evidence.
- Represent missing information explicitly rather than treating it as confirmation.
- Distinguish the presence of related information from sufficient evidence.
- Preserve the source supporting every extracted value.
- Separate extraction from subsequent interpretation and validation.
- Evaluate different failure categories independently rather than relying only on an aggregate performance score.

**What ReconIQ must develop independently**

- Trade matching and entity resolution.
- Cross-system data normalisation.
- Financial field comparison and tolerance rules.
- Transaction lifecycle and temporal reasoning.
- Source authority and evidence reliability.
- Exception classification and investigation workflows.

**8. Overall conclusion**

ROU-TransQA-v2 provides a methodological foundation for thinking about evidence reliability, answerability and abstention. These principles are relevant to ReconIQ because post-trade reconciliation also requires systems to distinguish supported conclusions from unsupported ones.

However, ReconIQ is not simply a contract question-answering system applied to financial records.

Its central challenge is broader: it must extract information from multiple sources, establish which records represent the same transaction, assess the sufficiency and consistency of evidence, and validate the resulting comparison.

**The strongest adaptation is therefore to use the paper's answerability-aware and evidence-grounded principles as the foundation of ReconIQ's evidence-handling layer, while developing separate financial-domain components for matching, validation and reconciliation.**

This preserves the original research contribution without attributing trade-reconciliation capabilities or experimental findings to a paper that did not study them.

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**PROMPT-10: I want to critically challenge the proposed connection between the research paper and ReconIQ. Is answerability-aware extraction sufficient for a financial reconciliation system? What additional mechanisms are needed to compare two trade records reliably? Explain the roles of deterministic comparison, candidate matching, duplicate detection and human review. Identify any weaknesses in the proposed research adaptation.**

**Critical assessment: Is answerability-aware extraction sufficient for ReconIQ?**

**No. Answerability-aware extraction is necessary for reliable evidence handling, but it is not sufficient to build a financial reconciliation system.**

The original research paper, ROU-TransQA-v2, addresses a specific problem: determining whether contractual evidence supports an answer and extracting that evidence when available.

ReconIQ faces a broader problem. It must determine whether two records represent the same financial transaction, compare their attributes, identify discrepancies and establish whether the available evidence justifies a reconciliation decision.

A system can extract every field correctly and still reconcile two trades incorrectly.

For example:

- Internal record: 100 shares, ₹500, 3 November.
- External record: 100 shares, ₹502, 3 November.

Extraction may be completely accurate. But ReconIQ must still determine whether these records represent the same trade and whether the price difference is legitimate or an exception.

This requires additional mechanisms.

**1. Why answerability-aware extraction is insufficient**

Consider four separate capabilities:

| **Capability** | **Question it answers** |
|---|---|
| Information extraction | What does each record contain? |
| Answerability assessment | Is the requested information supported by the source? |
| Record matching | Do these records represent the same trade? |
| Reconciliation validation | Do the matched records satisfy the required business rules? |

These capabilities are related but not interchangeable.

Answerability addresses evidence availability. It does not independently establish transaction identity, numerical consistency or reconciliation correctness.

For example, both records may contain a quantity of 100. That does not prove they represent the same transaction. They could be two separate purchases of the same security.

Therefore, ReconIQ needs a reconciliation architecture that goes beyond the original paper's QA framework.

**2. Additional mechanisms required**

**A. Deterministic comparison**

Deterministic comparison means applying explicitly defined rules to compare extracted values.

For example:

| Field | Internal | External | Comparison |
|---|---|---|---|
| Security ID | ISIN-A123 | ISIN-A123 | Match |
| Quantity | 100 | 100 | Match |
| Price | ₹500 | ₹502 | Mismatch |
| Currency | INR | INR | Match |
| Settlement date | 5 Nov | 5 Nov | Match |

The system can apply predefined rules:

- Security identifiers must match.
- Quantity must match, subject to defined allocation or partial-execution rules.
- Price must match within an approved tolerance, where applicable.
- Currency must match.
- Dates must be compared using the relevant date conventions.

**Why is this necessary?**

Financial reconciliation often involves exact numerical and structured comparisons. A language model should not decide whether ₹500 and ₹502 are equal or invent an acceptable tolerance.

Those decisions should be governed by explicit business rules.

There are also important complications:

- Different date formats may represent the same date.
- Prices may have different precision.
- Currency conversions may be involved.
- Fees may explain differences in net amounts.
- Partial executions may require aggregation.

Thus, deterministic comparison needs normalisation and domain-specific rules before it can produce reliable results.

**B. Candidate matching**

Before comparing fields, ReconIQ must identify which internal record corresponds to which external record.

This is called candidate matching or record linkage.

Consider:

**Internal records**

| Trade ID | Security | Quantity | Date |
|---|---|---|---|
| I-101 | ABC | 100 | 3 Nov |
| I-102 | ABC | 200 | 3 Nov |
| I-103 | XYZ | 50 | 4 Nov |

**External records**

| Reference | Security | Quantity | Date |
|---|---|---|---|
| B-501 | ABC | 100 | 3 Nov |
| B-502 | ABC | 200 | 3 Nov |
| B-503 | XYZ | 50 | 4 Nov |

Internal and external systems may use different identifiers.

ReconIQ therefore needs to identify candidate pairs using multiple attributes, such as:

- Security identifier.
- Trade date.
- Buy/sell direction.
- Quantity.
- Price.
- Broker.
- Account.
- Currency.
- Execution reference.

Candidate matching can use exact identifiers where available and more flexible matching rules where they are not.

For large datasets, it may first reduce the search space using blocking techniques, such as grouping records by account, date and security.

A matching algorithm can then rank candidate pairs based on attribute similarity.

**Why is this necessary?**

Without candidate matching, ReconIQ might compare unrelated trades simply because they share an amount or security.

It also needs to handle one-to-many relationships, such as one internal order corresponding to multiple broker executions.

**C. Duplicate detection**

Duplicate detection is another distinct requirement.

Suppose the internal OMS contains one trade:

| Trade | Quantity |
|---|---|

| | |
|---|---|
| I-101 | 100 |

But the external broker feed contains:

| Reference | Quantity |
|---|---|
| B-501 | 100 |
| B-501 | 100 |

The external feed may have delivered the same confirmation twice.

If ReconIQ processes both as separate transactions, it could incorrectly conclude that the external quantity is 200.

This can produce false reconciliation exceptions or incorrect position calculations.

Duplicate detection should consider:

- Repeated transaction references.
- Identical execution attributes.
- Repeated file ingestion.
- Corrections and replacement records.
- Partial executions that legitimately share an order reference.

The distinction between a duplicate and a legitimate repeated transaction is important.

For example, two executions of 50 shares under one order may be valid. They must not be discarded merely because their order reference is identical.

**Why is this necessary?**

Duplicate detection protects the integrity of the reconciliation population before matching and comparison take place.

It should be designed with idempotent ingestion and transaction-level identity rules.

**D. Human review**

Not every case can be resolved automatically.

Consider:

- Internal record says the trade settled.
- Custodian record says processing is complete.
- Broker confirmation contains no settlement status.

The system cannot safely conclude that these statuses mean the same thing.

A human reviewer may need to inspect the source documents, transaction history and relevant operational rules.

Human review is particularly important for:

- Ambiguous transaction matches.
- Conflicting records.
- Missing critical fields.
- Unusual price or quantity differences.
- Corporate actions and complex transaction structures.

- Cases requiring interpretation of business rules.

A useful system should provide reviewers with the evidence behind its decision, rather than simply presenting a warning.

For example:

**ReconIQ exception example**

Human review required

**Issue:** Settlement status cannot be verified.

| Source | Recorded status |
|---|---|
| Internal portfolio accounting | Settled |
| Custodian | Processed |
| Broker | Not provided |

**Reason for escalation**

The available statuses do not establish that the trade has completed settlement.

**Evidence available**

- Internal transaction record.
- Custodian processing record.
- Missing broker settlement confirmation.

**System decision:** Do not automatically mark the trade as fully reconciled.

Human review should also capture the reviewer's decision and supporting rationale. This can support auditability and improve future exception rules.

**3. How the four mechanisms work together**

These mechanisms should operate as distinct stages rather than being combined into one AI prediction.

**1. Data ingestion and normalisation**

Internal OMS, portfolio accounting, broker and custodian records

**2. Extraction and evidence handling**

Extract fields, preserve source references, identify missing and ambiguous information

**3. Duplicate detection**

Identify repeated records and distinguish duplicates from legitimate executions

**4. Candidate matching**

Identify corresponding internal and external trade records

**5. Deterministic validation**

Compare quantities, prices, dates, currencies and other attributes using defined rules

**6. Reconciliation decision**

Matched and consistent

Discrepancy or insufficient evidence

Unmatched or duplicate exception

**7. Human investigation**

Review unresolved cases and record decisions

The workflow is conceptual. In an actual implementation, some steps may operate iteratively. For example, human reviewers may identify a legitimate one-to-many match that requires the matching process to be rerun.

**4. Weaknesses in the proposed research adaptation**

The earlier proposal to adapt ROU-TransQA-v2 to ReconIQ has useful ideas, but several weaknesses need to be addressed before treating it as a credible research design.

| **Weakness** | **Why it matters** | **What is needed** |
|---|---|---|
| Overemphasis on answerability | Knowing that a field exists does not establish that two records match | Dedicated record-linkage mechanism |
| Binary answerability model | Financial records can be incomplete, conflicting, delayed or ambiguous | More granular evidence states |
| Insufficient treatment of structured data | RoBERTa extractive QA is not a complete solution for databases and numerical records | Structured-data processing and normalisation |
| No explicit duplicate handling | Duplicate records can distort reconciliation results | Deduplication and idempotent ingestion |
| No matching methodology | Similar values do not necessarily identify the same trade | Candidate generation and matching algorithms |
| No financial validation rules | Text extraction cannot determine acceptable price or quantity tolerances | Deterministic, configurable business rules |
| Weak treatment of time | Trade, posting and settlement dates represent different events | Temporal and lifecycle reasoning |
| No defined human-review mechanism | Ambiguous cases cannot always be resolved automatically | Exception queues, reviewer decisions and audit trails |
| No ReconIQ-specific evaluation | The paper's metrics do not measure trade-matching or reconciliation correctness | Financial-domain datasets and evaluation |

**A particularly important weakness: using QA metrics to evaluate reconciliation**

The original paper reports Exact Match, overall F1, HasAns F1 and NoAns F1.

These are relevant to its contractual QA task.

They would not be sufficient to evaluate ReconIQ.

ReconIQ would need additional measurements, such as:

| **Evaluation dimension** | **Possible measure** |
|---|---|

| | |
|---|---|
| Trade matching | Precision, recall and F1 of matched record pairs |
| Duplicate detection | Duplicate detection precision and recall |
| Field validation | Field-level mismatch detection accuracy |
| Exception detection | Precision and recall of exception identification |
| Evidence quality | Correct source attribution and evidence completeness |
| Automation | Percentage of cases resolved without human intervention |
| Operational quality | False reconciliation rate and unresolved exception rate |

These are proposed evaluation dimensions, not results reported by the original paper.

One particularly important metric would be the **false reconciliation rate**: the proportion of trades incorrectly marked as successfully reconciled.

In financial operations, a false successful match can be more damaging than an exception that is escalated for review.

**5. What should the research adaptation actually claim?**

A weak research claim would be:

"ROU-TransQA-v2 can be applied to post-trade reconciliation by using answerability-aware extraction."

That overstates the transferability of the original framework.

A stronger and more defensible claim would be:

"ROU-TransQA-v2 provides methodological inspiration for evidence-grounded extraction and answerability assessment in ReconIQ. However, post-trade reconciliation requires additional mechanisms for transaction matching, duplicate detection, deterministic field validation and human-supervised exception resolution."

This distinguishes the original contribution from the proposed extension.

**6. Final assessment**

**What answerability-aware extraction contributes**

- Prevents unsupported values from being presented as confirmed.
- Makes missing evidence explicit.
- Preserves the distinction between absent and contradictory information.
- Supports traceable evidence extraction.

**What it does not solve**

- Whether two records describe the same trade.
- Whether a record is duplicated.
- Whether numerical differences are acceptable.
- Whether transaction statuses are equivalent.
- Whether a trade should be marked reconciled.

**The central research insight is that answerability-aware extraction is an evidence-reliability component, not a complete reconciliation methodology.**

ReconIQ should combine it with candidate matching, duplicate detection, deterministic comparison and human review. The most defensible adaptation is therefore a hybrid architecture: AI-assisted evidence extraction and uncertainty assessment, combined with explicit financial rules and controlled reconciliation decisions.

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**PROMPT-11: Based on the research mapping, help me translate the proposed methodology into functional requirements for ReconIQ. The system compares internal and external financial trade records. It should consider: - Trade identity. - Trade date. - Buy/sell side. - Quantity. - Price and currency. - Settlement and allocation. Explain how the answerability concept should influence the handling of incomplete evidence. Separate research-inspired requirements from standard reconciliation requirements.**

**ReconIQ: Functional requirements for post-trade reconciliation**

ReconIQ should be designed as a financial reconciliation system with an evidence-aware decision layer.

Its primary function is to compare internal trade records against external broker or custodian records, identify corresponding transactions, validate their attributes and produce traceable reconciliation outcomes.

The research paper contributes the principles of answerability-aware extraction, evidence-grounded answering and abstention when evidence is insufficient. These principles should influence how ReconIQ handles missing or ambiguous information.

However, trade matching, field comparison, duplicate detection and settlement validation are standard reconciliation requirements. They should not be presented as contributions of the original paper.

The requirements below distinguish these two categories.

**1. Proposed system scope**

ReconIQ should process two broad categories of input:

- **Internal records:** OMS and portfolio accounting records.
- **External records:** Broker confirmations, custodian records and settlement or allocation reports.

The system should produce:

1. Matched trade records.
2. Field-level comparison results.
3. Missing or conflicting information alerts.
4. Settlement and allocation validation results.
5. Evidence references supporting each decision.
6. Exception cases requiring investigation.

**2. Core trade attributes and functional requirements**

Each trade should be represented using a common set of attributes. ReconIQ must preserve the original source values while creating normalised values for comparison.

| Attribute | Internal record | External record | Validation requirement |
|---|---|---|---|
| Trade identity | Internal trade/order ID | Broker or custodian reference | Establish whether records represent the same trade |
| Trade date | Internal execution date | External execution date | Compare dates using defined date conventions |
| Buy/sell side | Buy or sell | Buy or sell | Check directional consistency |

| | | | |
|---|---|---|---|
| Quantity | Executed quantity | Confirmed quantity | Compare, allowing for legitimate partial executions |
| Price | Internal execution price | External execution price | Compare using approved precision and tolerance |
| Currency | Trade currency | Confirmed currency | Validate currency consistency |
| Settlement | Settlement date and status | External settlement date and status | Compare dates and lifecycle states separately |
| Allocation | Account, portfolio or fund allocation | External allocation details | Verify allocation of the trade across relevant accounts |

**Functional requirements**

Standard reconciliation

**FR-01: Trade identity and candidate matching**

ReconIQ shall identify potential corresponding internal and external trade records using available identifiers and trade attributes.

- Use unique trade or execution references where available.
- Use combinations of security identifier, account, date, side, quantity and broker where direct identifiers differ.
- Support one-to-one and legitimate one-to-many matching.
- Avoid treating similarity in a single attribute as sufficient proof of identity.

**Expected result:** A candidate match, unmatched record or unresolved match.

Standard reconciliation

**FR-02: Trade date validation**

ReconIQ shall compare internal and external trade dates.

- Preserve the original dates.
- Normalise date formats.
- Distinguish trade date from settlement date and posting date.
- Flag differences for investigation where date conventions do not explain them.

**Expected result:** Date match, date discrepancy or insufficient date evidence.

Standard reconciliation

**FR-03: Buy/sell side validation**

ReconIQ shall compare the transaction direction.

- Buy must be compared with buy.
- Sell must be compared with sell.
- Missing or unclear side information must not be inferred from quantity or price.
- Opposite directions should trigger an exception.

**Expected result:** Side match, side mismatch or unavailable evidence.

Standard reconciliation

**FR-04: Quantity validation**

ReconIQ shall compare executed quantities across corresponding records.

- Support partial executions.
- Support aggregation of multiple executions where appropriate.
- Detect quantity differences.
- Avoid treating an order quantity as equivalent to an executed quantity without checking execution status.

**Expected result:** Quantity match, quantity discrepancy or unresolved quantity comparison.

Standard reconciliation

**FR-05: Price and currency validation**

ReconIQ shall compare execution prices and associated currencies.

- Compare prices only after currency and precision checks.
- Apply configured price tolerances where permitted.
- Avoid comparing nominal values across different currencies as though they were equivalent.
- Preserve original values and record any normalisation or conversion.

**Expected result:** Price and currency match, discrepancy or insufficient evidence.

Standard reconciliation

**FR-06: Settlement validation**

ReconIQ shall compare settlement dates and statuses.

- Distinguish settlement date from trade date.
- Distinguish pending, processed and settled states.
- Identify missing settlement confirmations.
- Avoid interpreting a processing status as proof of completed settlement unless the relevant business rules establish that equivalence.

**Expected result:** Settlement verified, settlement discrepancy or settlement unverified.

Standard reconciliation

**FR-07: Allocation validation**

ReconIQ shall compare trade allocations across internal and external records.

- Verify account, portfolio or fund identifiers.
- Support allocation splits.
- Compare allocated quantities against the total executed quantity.
- Identify unallocated, overallocated and incorrectly allocated quantities.
- Support one trade being distributed across multiple accounts.

**Expected result:** Allocation consistent, allocation discrepancy or allocation incomplete.

**3. Research-inspired requirements: answerability and evidence handling**

These requirements translate the methodological principles of ROU-TransQA-v2 into the proposed ReconIQ design. They are adaptations, not features demonstrated by the original paper in a financial reconciliation environment.

Research-inspired

**FR-08: Evidence-supported field extraction**

ReconIQ shall extract trade attributes only from information available in the relevant source records.

For each extracted value, the system should retain:

- Source system.
- Source record or document identifier.
- Original field value.
- Normalised field value, if applicable.
- Relevant timestamp or document location.

Example:

| **Field** | **Value** | **Source** |
|---|---|---|
| Quantity | 100 | Internal OMS |
| Quantity | 100 | Broker confirmation |
| Settlement date | 5 Nov | Internal OMS |
| Settlement date | Not provided | Broker confirmation |

The system must not copy the internal settlement date into the external record.

Research-inspired

**FR-09: Answerability-aware evidence assessment**

ReconIQ shall determine whether the available source evidence is sufficient to support a requested field-level conclusion.

It should distinguish:

- Supported: the source explicitly provides the required information.
- Not provided: the source does not contain the required information.
- Ambiguous: the source contains related information but does not establish the required fact.
- Conflicting: available sources provide incompatible values.
- Not applicable: the field is not relevant to that transaction or source.

These are proposed ReconIQ evidence states. They are not the original paper's formal classification scheme.

Research-inspired

**FR-10: Abstention when evidence is insufficient**

ReconIQ shall not automatically confirm a field or trade when supporting evidence is absent or insufficient.

For example:

| **Internal settlement date** | **External settlement date** | **Required result** |
|---|---|---|
| 5 Nov | 5 Nov | Agreement |

| | | |
|---|---|---|
| 5 Nov | 7 Nov | Disagreement |
| 5 Nov | Missing | Insufficient evidence |
| Missing | Missing | Insufficient evidence |
| 5 Nov | Ambiguous status only | Unresolved |

The system should preserve the available value while withholding an unsupported conclusion.

Research-inspired

**FR-11: Evidence traceability**

ReconIQ shall provide supporting source references for each field-level comparison and final reconciliation decision.

A reviewer should be able to identify:

- Which internal record supplied the value.
- Which external record supplied the comparison value.
- Which values were compared.
- Which rule was applied.
- Why the result was classified as a match, discrepancy or unresolved case.

Research-inspired

**FR-12: Separation of extraction and validation**

ReconIQ shall keep the extraction process separate from the validation process.

The extraction component must report what the source says. The validation component must determine whether the extracted values satisfy reconciliation rules.

For example:

- Extraction: Internal quantity = 100; external quantity = 90.
- Validation: Quantity discrepancy = 10.
- Decision: Exception requiring investigation.

The extraction component must not silently modify 90 to 100 or infer that the difference is acceptable.

**4. Additional standard reconciliation requirements**

These requirements are essential to the financial reconciliation workflow but are not derived directly from the research paper.

Standard reconciliation

**FR-13: Data normalisation**

ReconIQ shall standardise incoming records before comparison.

It should handle:

- Date and timestamp formats.
- Currency codes.
- Decimal precision.
- Security identifiers.

- Buy/sell representations.
- Account and portfolio identifiers.
- Units of quantity.

Original source values must remain available for audit purposes.

Standard reconciliation

**FR-14: Duplicate detection**

ReconIQ shall identify duplicate records before reconciliation.

It should:

- Detect repeated transaction references.
- Identify duplicate file ingestion.
- Distinguish duplicate confirmations from legitimate partial executions.
- Prevent duplicate records from being counted as separate trades.
- Retain duplicate detection evidence.

Standard reconciliation

**FR-15: Configurable validation rules**

ReconIQ shall support configurable comparison rules.

Examples:

- Exact quantity matching.
- Approved price tolerances.
- Currency consistency.
- Permitted date differences.
- Allocation totals matching executed quantity.

Rules should be configurable by relevant transaction type, asset class or operational requirement. They should not be hardcoded as universal assumptions.

Standard reconciliation

**FR-16: Exception management and human review**

ReconIQ shall create exceptions for records that cannot be reconciled automatically.

The exception should include:

- Trade reference.
- Affected field.
- Internal and external values.
- Evidence availability.
- Reason for the exception.
- Relevant source records.
- Review status.

Reviewers should be able to record a resolution and supporting explanation. The system should retain the history of the decision.

Standard reconciliation

**FR-17: Reconciliation status management**

ReconIQ shall maintain separate statuses for trade matching, field validation and overall reconciliation.

For example:

| **Dimension** | **Status** |
|---|---|
| Trade identity | Matched |
| Trade date | Matched |
| Buy/sell side | Matched |
| Quantity | Matched |
| Price | Discrepancy |
| Currency | Matched |
| Settlement | Insufficient evidence |
| Allocation | Matched |
| Overall reconciliation | Exception |

This is important because a trade can be correctly matched while still having unresolved discrepancies.

**5. How answerability should influence incomplete evidence**

This is the most important research-inspired requirement.

ReconIQ should not interpret missing information as either agreement or disagreement.

Consider this example:

**Trade settlement verification**

| **Attribute** | **Internal OMS** | **External broker** |
|---|---|---|
| Trade identity | TR-1025 | TR-1025 |
| Trade date | 3 Nov | 3 Nov |
| Side | Buy | Buy |
| Quantity | 100 | 100 |
| Price | ₹500 | ₹500 |

| | | |
|---|---|---|
| Currency | INR | INR |
| Settlement date | 5 Nov | Missing |

Insufficient evidence

**ReconIQ decision**

The trade is matched on its available identifying attributes. However, the external confirmation does not establish the settlement date.

The system must not mark external settlement-date verification as successful.

There are three distinct concepts here:

1. **Trade identity:** The system has identified the corresponding external trade.
2. **Field agreement:** The available values agree.
3. **Evidence sufficiency:** The available evidence is adequate to support the conclusion.

These must be evaluated independently.

A missing settlement date does not necessarily mean the entire trade is unmatched. It means that settlement-date verification remains incomplete.

The system should also consider whether a particular source is expected to contain that field. For example, a broker execution confirmation may not be the authoritative source for final settlement status. In that case, the custodian record may be the appropriate source for verification.

**6. Consolidated requirements matrix**

Research-inspired requirements are adapted from the paper's methodology. Standard requirements arise from the operational needs of trade reconciliation.

| ID | Functional requirement | Classification |
|---|---|---|
| FR-01 | Trade identity and candidate matching | Standard |
| FR-02 | Trade date validation | Standard |
| FR-03 | Buy/sell side validation | Standard |
| FR-04 | Quantity validation | Standard |
| FR-05 | Price and currency validation | Standard |
| FR-06 | Settlement validation | Standard |
| FR-07 | Allocation validation | Standard |
| FR-08 | Evidence-supported extraction | Research-inspired |
| FR-09 | Answerability-aware evidence assessment | Research-inspired |

| | | |
|---|---|---|
| FR-10 | Abstention for insufficient evidence | Research-inspired |
| FR-11 | Evidence traceability | Research-inspired |
| FR-12 | Separation of extraction and validation | Research-inspired |
| FR-13 | Data normalisation | Standard |
| FR-14 | Duplicate detection | Standard |
| FR-15 | Configurable validation rules | Standard |
| FR-16 | Exception management and human review | Standard |
| FR-17 | Reconciliation status management | Standard |

**7. Recommended processing logic**

The functional requirements can be organised into this logical sequence:

1. **Ingest records**

Collect internal and external trade records.

2. **Normalise and detect duplicates**

Standardise fields and remove or flag duplicate records.

3. **Extract and preserve evidence**

Identify trade attributes and their source references.

4. **Assess evidence sufficiency**

Identify supported, missing, ambiguous and conflicting information.

5. **Match trade records**

Identify corresponding internal and external transactions.

6. **Validate trade attributes**

Compare date, side, quantity, price, currency, settlement and allocation.

7. **Generate reconciliation outcome**

Produce matched, exception, incomplete or unresolved results.

8. **Escalate unresolved cases**

Provide evidence and comparison details for human review.

**8. How to position the research contribution**

The distinction should be explicit in any ReconIQ research documentation.

**Research-inspired contribution**

- Answerability-aware evidence handling.
- Explicit representation of missing information.
- Abstention instead of unsupported confirmation.

- Evidence-grounded extraction.
- Separation of extraction and downstream validation.

**Financial reconciliation contribution**

- Trade matching.
- Duplicate detection.
- Numerical and date comparison.
- Settlement and allocation validation.
- Configurable financial rules.
- Exception management and human review.

The original paper motivates the evidence-handling layer. The reconciliation requirements provide the financial-domain functionality.

**Overall design principle:** ReconIQ should never equate successful extraction with successful reconciliation. It must establish trade identity, compare the relevant attributes, verify that the evidence is sufficient and only then determine whether a trade can be marked reconciled.

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**PROMPT-12: I want to understand how candidate matching and final reconciliation should be separated. Compare exact matching, fuzzy matching and hybrid matching for trade records. Why should similarity scores not independently confirm a trade match? How should duplicates, competing candidates, missing values and conflicting fields be handled? Explain which decisions should be deterministic and which may use AI assistance.**

**Candidate matching vs. final reconciliation in ReconIQ**

The key distinction is:

**Candidate matching determines which external record might represent the same trade. Final reconciliation determines whether the matched records agree sufficiently across the required financial attributes.**

These are separate decisions. A system can identify the correct corresponding trade and still discover that its price, quantity, settlement or allocation is incorrect.

For example:

| Attribute | Internal OMS | Broker record |
|---|---|---|
| Trade reference | INT-1025 | BRK-8891 |
| Security | ABC Ltd | ABC Ltd |
| Trade date | 3 Nov | 3 Nov |
| Side | Buy | Buy |
| Quantity | 100 | 100 |
| Price | ₹500 | ₹502 |

The records may represent the same trade, despite having different references. Candidate matching should identify the relationship. Final reconciliation should then flag the ₹2 price difference.

**1. Exact, fuzzy and hybrid matching**

**A. Exact matching**

Exact matching identifies records using equality across specified identifiers or attributes.

Example:

| Field | Internal | External | Result |
|---|---|---|---|
| Trade reference | TR-1025 | TR-1025 | Exact |
| Security ID | ISIN-A | ISIN-A | Exact |
| Account ID | ACC-10 | ACC-10 | Exact |

**Advantages**

- Fast and deterministic.
- Easy to explain and audit.
- Low risk of accidental matches when identifiers are unique and reliable.

**Limitations**

- Internal and external systems may use different reference numbers.
- Formatting differences can prevent matches.
- One trade may have multiple execution records.
- Missing identifiers can prevent matching even when records correspond.

Exact matching is appropriate when a shared identifier is authoritative, unique and correctly populated.

**B. Fuzzy matching**

Fuzzy matching identifies records based on similarity rather than exact equality.

It can compare:

- Security descriptions.
- Dates within an approved window.
- Quantities.
- Prices.
- Broker references.
- Account identifiers.
- Other transaction attributes.

Example:

| Field | Internal | External |
|---|---|---|
| Security | ABC Limited | ABC Ltd. |
| Trade date | 3 Nov | 3 Nov |

| | | |
|---|---|---|
| Side | Buy | Buy |
| Quantity | 100 | 100 |
| Price | ₹500 | ₹500.01 |

A fuzzy matcher may identify these records as highly similar.

**Advantages**

- Handles formatting differences.
- Can identify corresponding records without shared identifiers.
- Useful when external systems have inconsistent naming conventions.

**Limitations**

- Similarity does not establish transaction identity.
- Common securities can generate competing candidates.
- Similar values can occur in unrelated trades.
- Tolerances can hide genuine discrepancies if poorly configured.

Fuzzy matching should generate potential matches, not independently approve them.

**C. Hybrid matching**

Hybrid matching combines exact rules with fuzzy candidate generation.

A typical approach:

1. Match using reliable unique identifiers where available.
2. For unmatched records, generate candidates using selected attributes.
3. Score and rank the candidates.
4. Apply mandatory eligibility rules.
5. Resolve competing candidates.
6. Send uncertain cases for review.

Example:

Internal record:

- Security: ABC Ltd
- Trade date: 3 Nov
- Side: Buy
- Quantity: 100
- Price: ₹500
- Internal reference: INT-1025

External candidates:

| Candidate | Security | Date | Side | Quantity | Price |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| A | ABC Ltd | 3 Nov | Buy | 100 | ₹500 |
| B | ABC Ltd | 3 Nov | Buy | 100 | ₹500 |
| C | ABC Ltd | 3 Nov | Sell | 100 | ₹500 |

Candidates A and B are indistinguishable using the displayed attributes. Candidate C has an opposite side and should be rejected.

The system should not arbitrarily select A or B. It needs another identifying attribute, such as execution reference, account, broker or timestamp.

Recommended approach

**Hybrid matching**

It provides the strongest practical structure because it combines the reliability of exact identifiers with the flexibility of fuzzy candidate discovery.

**2. Why similarity scores cannot independently confirm a match**

A similarity score measures how closely two records resemble each other. It does not prove that they represent the same economic transaction.

There are four important reasons.

**1. Different trades can have identical attributes.**

Two investors can buy the same security, on the same day, at the same price and quantity.

**2. A high score can hide a critical mismatch.**

Consider two records that match on security, date, quantity and price but have opposite buy/sell sides.

A weighted average could still produce a high similarity score. Yet these records cannot be treated as the same transaction under ordinary matching rules.

**3. Missing information can artificially increase confidence.**

If the external record omits the account identifier, the matcher must not interpret the missing value as agreement.

**4. Similarity does not establish source authority.**

A broker execution confirmation and a custodian settlement record may contain different information because they describe different stages of the transaction.

A score cannot determine whether a particular source is authoritative for a particular field.

**Better decision structure**

Use similarity scores for ranking candidates, then apply mandatory rules and identity checks.

For example:

| **Matching condition** | **Decision** |
|---|---|
| Unique authoritative reference agrees | Strong candidate |
| Reference unavailable, other attributes agree | Candidate requiring further validation |
| Opposite buy/sell side | Reject candidate |

| | |
|---|---|
| Multiple equally plausible candidates | Ambiguous; do not auto-match |
| High score but critical identifier conflicts | Reject or investigate |

A score threshold alone should never be the final acceptance condition.

**3. Handling difficult matching situations**

**A. Duplicates**

Duplicates occur when the same trade is received multiple times, possibly through repeated file ingestion or repeated external confirmations.

Example:

Internal record: 100 shares of ABC Ltd.

External records:

- Broker confirmation A: 100 shares.
- Broker confirmation B: 100 shares, same execution reference.

The system must not match both records independently to the same internal trade.

Required handling:

1. Detect duplicate ingestion using source identifiers and record fingerprints.
2. Preserve the original records and their provenance.
3. Determine whether the records are true duplicates or legitimate separate executions.
4. Prevent duplicate records from inflating reconciled quantities.
5. Flag uncertain cases for investigation.

**Important distinction:**

One order executed in two parts:

- Execution A: 60 shares.
- Execution B: 40 shares.

These are not duplicates merely because they belong to the same order.

ReconIQ should support one-to-many matching and aggregation where the business rules permit it.

**B. Competing candidates**

Suppose one internal trade has three possible external matches.

The system should:

1. Generate eligible candidates.
2. Rank them using available attributes.
3. Check mandatory identity constraints.
4. Determine whether one candidate is sufficiently distinguishable.
5. Abstain from automatic matching if multiple candidates remain plausible.

Example:

| Candidate | Security | Date | Side | Quantity | Assessment |
|---|---|---|---|---|---|
| A | ABC | 3 Nov | Buy | 100 | Plausible |
| B | ABC | 3 Nov | Buy | 100 | Plausible |
| C | ABC | 3 Nov | Sell | 100 | Rejected |

A and B cannot be distinguished from the supplied information.

The appropriate result is **ambiguous match**, not an arbitrary selection.

The system should also consider global matching constraints: one external execution should not be assigned to multiple unrelated internal trades unless the transaction structure explicitly permits it.

**C. Missing values**

Missing information should be treated as unknown, not as a match or mismatch.

Example:

| Field | Internal | External | Result |
|---|---|---|---|
| Security | ABC | ABC | Agreement |
| Date | 3 Nov | 3 Nov | Agreement |
| Quantity | 100 | 100 | Agreement |
| Account | ACC-10 | Missing | Unverified |

The system can retain the candidate match while marking account validation as incomplete.

This is where the paper's answerability principle becomes useful: when the evidence cannot support a conclusion, the system should abstain rather than fabricate or infer a value.

However, missingness should be classified further:

- Not provided by source.
- Not yet received.
- Not applicable.
- Unavailable due to ingestion failure.
- Expected but missing.

These cases have different operational implications.

**D. Conflicting fields**

Conflicting fields are different from missing fields.

Example:

| Field | Internal | External | Result |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Security | ABC | ABC | Agreement |
| Date | 3 Nov | 3 Nov | Agreement |
| Side | Buy | Buy | Agreement |
| Quantity | 100 | 100 | Agreement |
| Price | ₹500 | ₹502 | Conflict |

The system may establish that both records represent the same trade but must flag the price discrepancy.

It should not change the external value to match the internal value.

Each field should be evaluated independently using the relevant validation rules.

**4. Which decisions should be deterministic and which can use AI?**

This separation is central to a reliable ReconIQ architecture.

| **System activity** | **Preferred mechanism** | **Reason** |
|---|---|---|
| Field extraction from structured records | Deterministic parsing; AI for unstructured inputs | Structured values should be read directly where possible |
| Date and currency normalisation | Deterministic | Reproducible transformations |
| Exact identifier comparison | Deterministic | Equality is objectively testable |
| Candidate generation | Hybrid | Rules and AI-assisted retrieval can find possible matches |
| Fuzzy similarity scoring | Statistical/AI-assisted | Useful for ranking uncertain candidates |
| Duplicate detection | Deterministic rules plus statistical detection | Exact references catch known duplicates; similarity helps find near-duplicates |
| Quantity comparison | Deterministic | Numerical comparison |
| Price tolerance checks | Deterministic | Governed by approved business rules |
| Buy/sell validation | Deterministic | Critical transaction constraint |
| Settlement status mapping | Deterministic mapping rules; AI may assist interpretation | Status meanings must be controlled |
| Allocation validation | Deterministic | Requires accounting and quantity constraints |

| | | |
|---|---|---|
| Evidence sufficiency assessment | Rules plus AI assistance | AI can interpret unstructured evidence, but must not invent missing values |
| Exception prioritisation | AI-assisted | Can help identify patterns and prioritise investigation |
| Final approval of uncertain cases | Human review | Requires accountable operational judgement |

**Appropriate uses of AI**

AI can assist with:

- Extracting trade fields from broker confirmation PDFs.
- Recognising different representations of security names.
- Generating possible matching candidates.
- Identifying semantically similar status descriptions.
- Explaining why records were flagged.
- Summarising evidence for reviewers.
- Prioritising unusual exceptions.

AI should not independently:

- Override a conflicting trade reference.
- Change a quantity or price.
- Declare a missing settlement date verified.
- Ignore a buy/sell mismatch.
- Resolve competing candidates without sufficient identity evidence.
- Mark a financially significant exception as reconciled solely based on a confidence score.

The final reconciliation decision should be governed by explicit rules and controlled exception handling.

**5. Recommended ReconIQ architecture**

**1. Record ingestion**

Internal OMS and portfolio accounting records; external broker and custodian records

**2. Normalisation and duplicate detection**

Deterministic rules and duplicate detection logic

**3. Candidate matching**

Exact matching + fuzzy matching + hybrid ranking

**4. Identity decision**

Unique match, ambiguous match, unmatched or rejected candidate

**5. Field-level reconciliation**

Deterministic comparison of date, side, quantity, price, currency, settlement and allocation

**6. Evidence sufficiency assessment**

Supported, missing, ambiguous or conflicting evidence

**7. Reconciliation outcome**

Reconciled, exception, incomplete or unresolved

**8. Human review and audit trail**

Review uncertain matches, resolve exceptions and record decisions

**6. The most important architectural distinction**

ReconIQ should maintain two independent outputs:

**Output A: Matching decision**

- Matched.
- Unmatched.
- Ambiguous.
- Rejected.

**Output B: Reconciliation decision**

- Reconciled.
- Discrepancy.
- Incomplete evidence.
- Unresolved.

This prevents a common design error: treating a high-confidence record match as proof that the underlying financial data agrees.

For example:

**Trade identity**

Matched

**Field validation**

Discrepancy

**Final result: Exception**

The records represent the same trade, but their execution prices differ. The system should preserve the match and raise a price exception.

**Design conclusion:** Use AI and similarity methods to discover and interpret possible relationships. Use deterministic rules to validate financial facts. Use answerability-aware evidence handling to prevent unsupported conclusions. Use human review when identity or evidence remains uncertain.

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**PROMPT-13: Using the research paper, "An Answerability-Aware Contract Intelligence Framework for Right-of-Use Asset Management With IFRS 16-Oriented Indicator Mapping and Multidimensional Analytics", explain how its answerability-aware approach can be adapted to the evidence extraction component of ReconIQ. Do not assume that the paper directly studies financial trade reconciliation. Clearly distinguish the paper's original methodology from the adaptations proposed for ReconIQ. Structure your analysis as follows: 1. Research principle: Explain how the paper determines whether a question can be answered from available evidence. Explain why answerability should be assessed before attempting information extraction. 2. Application to ReconIQ: Explain how this principle applies when extracting trade details from broker confirmations, custodian statements and other external financial records. 3. Extraction outcomes: Define four possible outcomes: - ANSWERABLE: the required information is explicitly supported by the evidence. - NOT_FOUND: the evidence does not contain the required information. - AMBIGUOUS: the evidence contains conflicting or multiple possible interpretations. - INVALID_EVIDENCE: the supplied evidence is unreadable, malformed or otherwise unsuitable. 4. Evidence**

**traceability: Explain why every extracted value should retain its original evidence quote and source reference. Explain how this makes the result auditable. 5. Missing information: Use a settlement-date example. If the internal system contains a settlement date but the external document does not, explain why the system must preserve the unknown status rather than assume that the dates agree. 6. Human oversight: Explain which extraction outcomes should be escalated for human review and why an automated system should not invent or infer unsupported financial values. 7. Technical design: Propose a structured output for each extracted field containing: - field name - extracted value - extraction status - exact evidence quote - source document - confidence or validation information, where justified 8. Research connection: Create a table with three columns: - Principle from the research paper - How ReconIQ adapts it - What is different between the two applications 9. Limitations: Identify what cannot be claimed from the research paper and what would need independent testing in ReconIQ. 10. Final deliverable: Produce a concise technical design specification for the ReconIQ evidence extraction component. Do not write implementation code yet. Use precise academic language. Do not invent experimental results, claim that the research paper validates ReconIQ, or present proposed design decisions as findings from the paper.**

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**Adapting the answerability-aware methodology to ReconIQ's evidence extraction component**

**1. Research principle**

The paper, *An Answerability-Aware Contract Intelligence Framework for Right-of-Use Asset Management With IFRS 16-Oriented Indicator Mapping and Multidimensional Analytics*, introduces ROU-TransQA-v2, a framework for extracting answers from contractual evidence while explicitly accounting for whether the available evidence supports an answer.

Its central principle is that **a system should establish whether an answer is supported by the supplied evidence rather than assuming that every question must produce an extracted value.**

The framework combines:

- RoBERTa-based extractive question answering.
- Explicit answerability classification.
- Conditional masking of span supervision for unanswerable examples.
- Expert-informed mapping of extracted evidence to IFRS 16-oriented indicators.

The paper uses CUAD, a legal contract question-answering benchmark. Its enterprise validation also evaluates contract-related questions and IFRS 16-oriented indicator mapping. It does not study post-trade financial reconciliation.

IEEE Xplore

**Why answerability matters before extraction**

Consider a question asked of a contract:

*What is the lessee's termination notice period?*

There are two possibilities:

1. The contract explicitly states the notice period. The system has evidence from which it can extract an answer.
2. The contract does not state the notice period. The system should not manufacture an answer.

Without answerability awareness, an extraction system may return a plausible-looking value even when the source does not support it.

The paper addresses this problem through explicit answerability classification and conditional masking of span supervision for unanswerable examples. In practical terms, an unanswerable example should not be forced to produce an answer span during training.

The important distinction is between:

- **Answer extraction:** identifying the information expressed in the evidence.

- **Answerability assessment:** determining whether the evidence supports providing that information.

Answerability assessment prevents extraction from being treated as mandatory when the evidence is insufficient.

For ReconIQ, this principle is particularly relevant because an incorrect financial value can lead to an incorrect reconciliation decision.

**2. Application to ReconIQ**

ReconIQ receives trade information from multiple internal and external sources, including:

- Broker confirmations.
- Custodian statements.
- Settlement reports.
- Allocation reports.
- Other external transaction records.

Its evidence extraction component must retrieve relevant trade attributes from these sources.

For example, consider the requirement:

*Extract the settlement date for trade TR-1025 from the broker confirmation.*

The external document may contain:

- Trade reference: TR-1025.
- Security: ABC Ltd.
- Trade date: 3 November 2026.
- Quantity: 100.
- Price: ₹500.
- No settlement date.

A conventional extraction system might return a blank field, attempt to infer the settlement date, or copy the date from an internal system.

An answerability-aware component should instead determine whether the broker confirmation contains sufficient evidence for the requested field.

The proposed ReconIQ process is:

1. Identify the requested trade and field.
2. Locate the relevant source document or record.
3. Assess whether the required information is present and interpretable.
4. Extract the value only when supported by evidence.
5. Record the extraction outcome and supporting evidence.
6. Abstain when the evidence cannot support a reliable value.

This is an adaptation of the paper's evidence-grounded answerability principle. ReconIQ would apply it to structured financial attributes and financial documents rather than contractual questions.

A further distinction is necessary: **the absence of a field from one source does not establish that the information is unavailable across all sources.** The system may need to consult the appropriate authoritative source for that attribute. For example, a custodian settlement report may be more suitable for verifying settlement status than a broker execution confirmation.

**3. Extraction outcomes**

The following four outcomes are proposed for ReconIQ. They are an extension of the paper's answerability principle, not the paper's original four-class classification scheme.

| Outcome | Definition | ReconIQ example |
|---|---|---|
| ANSWERABLE | Required information is explicitly supported by the evidence. | Broker confirmation explicitly states quantity = 100. |
| NOT_FOUND | The evidence does not contain the required information. | Settlement date is absent from the broker confirmation. |
| AMBIGUOUS | Evidence contains conflicting values or multiple possible interpretations. | Two different settlement dates appear without clarification. |
| INVALID_EVIDENCE | Evidence is unreadable, malformed, corrupted or otherwise unsuitable for reliable extraction. | A scanned confirmation is too distorted to identify its trade details. |

**Important distinctions between the outcomes**

**ANSWERABLE**

The system can extract the value and attach supporting evidence.

**NOT_FOUND**

The field is absent from the examined evidence. The system must not infer the value from another record or from general assumptions.

**AMBIGUOUS**

Information exists, but the system cannot determine which value or interpretation is correct. For example, a document contains two execution quantities without explaining whether they represent separate fills or a correction.

**INVALID_EVIDENCE**

The system cannot reliably interpret the supplied evidence. This is different from NOT_FOUND: the information may be present, but the source cannot be read or processed reliably.

These statuses should remain distinct because they imply different operational actions.

For example, NOT_FOUND may require another source, while INVALID_EVIDENCE may require document replacement or manual inspection.

The four statuses are proposed as field-level extraction outcomes. They should not be treated as direct equivalents of the paper's original answerable/unanswerable labels.

**4. Evidence traceability**

Every extracted value should retain its original evidence quote and source reference.

A value without evidence is difficult to verify, especially when several systems provide different versions of the same transaction information.

Consider:

| Field | Extracted value | Evidence |
|---|---|---|
| Trade date | 03-Nov-2026 | "Trade Date: 03-Nov-2026" |

| | | |
|---|---|---|
| Quantity | 100 | "Executed Quantity: 100" |
| Price | 500.00 | "Execution Price: INR 500.00" |
| Currency | INR | "Currency: INR" |

Each value should also retain:

- Source document identifier.
- Source system.
- Page number or record location.
- Original text or field representation.
- Extraction timestamp.
- Normalisation details, if applicable.

**Why this is important**

**1. Auditability**

A reviewer can verify that the extracted value actually appears in the source.

**2. Error investigation**

If the quantity is incorrect, the reviewer can inspect the exact source passage instead of investigating the entire document.

**3. Source accountability**

If two documents provide different values, the system can identify which source supplied each value.

**4. Reproducibility**

A reviewer can reconstruct how the extraction component arrived at its output.

**5. Separation of evidence and interpretation**

The system preserves what the document says independently of what ReconIQ concludes about the trade.

This distinction is fundamental. An evidence quote establishes what was extracted; it does not automatically establish that the extracted value is financially correct.

**5. Missing information: settlement-date example**

Consider an internal and external trade record.

**Internal OMS record**

| **Field** | **Value** |
|---|---|
| Trade reference | TR-1025 |
| Trade date | 03-Nov-2026 |
| Settlement date | 05-Nov-2026 |

**External broker confirmation**

| **Field** | **Value** |
|---|---|

| | |
|---|---|
| Trade reference | TR-1025 |
| Trade date | 03-Nov-2026 |
| Settlement date | Not present |

NOT_FOUND

**ReconIQ extraction result**

External settlement date: unknown from this document.

The internal system contains 5 November, but the external document does not provide a settlement date.

ReconIQ must preserve the distinction between:

- Internal settlement date: 5 November.
- External settlement date: not found.
- Settlement-date comparison: not verifiable using the available external evidence.

It must not:

- Copy 5 November into the external settlement-date field.
- Treat the missing value as agreement.
- Treat the missing value as disagreement.
- Mark settlement as verified.

The correct conclusion is that the external settlement date is unverified from the supplied document.

This does not necessarily mean the trade itself is unmatched. The trade may be matched on its reference, date, side and other attributes while settlement verification remains incomplete.

The system may subsequently consult a custodian statement or settlement report if that is the appropriate source.

This is where the answerability-aware principle is valuable: **absence of evidence should result in abstention, not an invented financial value or an unsupported reconciliation conclusion.**

**6. Human oversight**

Human review should be driven by the extraction outcome and the financial importance of the field.

| **Extraction outcome** | **Recommended handling** | **Reason** |
|---|---|---|
| ANSWERABLE | Automated processing, subject to validation rules | Evidence supports extraction |
| NOT_FOUND | Review or retrieve another source when the field is required | Information is unavailable in the examined source |
| AMBIGUOUS | Escalate for human review | The system cannot establish the correct interpretation |
| INVALID_EVIDENCE | Escalate for document correction or manual extraction | Reliable automated interpretation is not possible |

Not every NOT_FOUND case necessarily requires immediate human intervention. If the source is not expected to contain that field, ReconIQ can mark it as unavailable from that source and seek the appropriate authoritative record.

However, missing mandatory evidence should prevent the relevant verification from being marked complete.

**Why unsupported inference must be prohibited**

Suppose the system has observed that a particular settlement date frequently follows a certain trade date.

It might predict 5 November based on historical patterns.

But a predicted settlement date is not evidence that the specific transaction was scheduled or settled on that date.

Similarly:

- A missing quantity must not be calculated from unrelated records without a justified business rule.
- A missing price must not be copied from a similar transaction.
- A missing allocation must not be inferred from an account's historical behaviour.
- A missing settlement status must not be interpreted as settled.

AI may assist in locating evidence, extracting values or identifying possible interpretations. It should not convert unsupported predictions into verified financial facts.

**7. Technical design: structured extraction output**

ReconIQ should generate a structured result for every requested trade attribute.

The proposed schema contains seven elements:

| **Element** | **Purpose** |
|---|---|
| field_name | Identifies the requested trade attribute |
| extracted_value | Extracted value, or null if unavailable |
| extraction_status | One of the four proposed outcomes |
| evidence_quote | Exact text supporting the extracted value |
| source_document | Document or record identifier and source system |
| confidence | Model confidence, if the extraction method provides a meaningful calibrated score |
| validation_information | Extraction checks, normalisation details or warnings |

**Example structured output**

The following is an illustrative design, not an observed output from the paper.


```
{
  "trade_reference": "TR-1025",
  "source_document": {
    "document_id": "BROKER-CONF-1025",
    "source_system": "Broker"
```

```
},
"extracted_fields": [
 {
  "field_name": "trade_date",
  "extracted_value": "2026-11-03",
  "extraction_status": "ANSWERABLE",
  "evidence_quote": "Trade Date: 03-Nov-2026",
  "source_document": "BROKER-CONF-1025",
  "confidence": null,
  "validation_information": {
   "normalised": true,
   "original_value": "03-Nov-2026"
  }
 },
 {
  "field_name": "quantity",
  "extracted_value": 100,
  "extraction_status": "ANSWERABLE",
  "evidence_quote": "Executed Quantity: 100",
  "source_document": "BROKER-CONF-1025",
  "confidence": null,
  "validation_information": {
   "unit": "shares"
  }
 },
 {
  "field_name": "settlement_date",
  "extracted_value": null,
  "extraction_status": "NOT_FOUND",
  "evidence_quote": null,
  "source_document": "BROKER-CONF-1025",
  "confidence": null,
  "validation_information": {
   "reason": "Field not present in supplied document"
  }
 }
```

]

}

**Design considerations**

**Confidence should not be confused with evidence sufficiency.**

A model may be highly confident in an incorrectly extracted value. Therefore:

- Confidence should be retained only when the extraction method provides a meaningful score.
- Confidence should be calibrated against labelled data before it is used for automated decision thresholds.
- An ANSWERABLE result must still have supporting evidence.
- A high confidence score must not override a missing or conflicting source value.

The source_document should ideally contain a stable document ID, source system, page number, section or record location, and document version.

For AMBIGUOUS, the design should retain all relevant competing evidence quotes rather than selecting one arbitrarily. For INVALID_EVIDENCE, it should record the reason for failure, such as unreadable page, corrupted file or malformed record.

**8. Research connection**

The following table separates the paper's original methodological principles from the proposed ReconIQ adaptations.

| **Principle from the research paper** | **How ReconIQ adapts it** | **What is different between the two applications** |
|---|---|---|
| Explicit answerability classification | Determine whether each requested trade attribute is supported by the source | Paper addresses contractual questions; ReconIQ extracts structured financial attributes |
| Extractive question answering | Extract values from broker confirmations, custodian statements and other records | ReconIQ must handle tables, structured fields, PDFs and transaction records, not only contractual text |
| Conditional masking of span supervision for unanswerable examples | Avoid training the extraction model to produce unsupported values when evidence is absent | ReconIQ requires financial-domain annotations and potentially different training objectives |
| Evidence-grounded extraction | Retain original evidence quotes and source references for each extracted field | ReconIQ needs field-level provenance across multiple financial systems |
| Answerability-aware abstention | Return NOT_FOUND or other unresolved outcomes instead of inventing values | ReconIQ proposes four extraction outcomes, extending beyond the paper's confirmed answerability classification |
| Expert-informed D1–D6 indicator mapping | Potentially map extracted trade attributes to financial validation requirements | The paper's D1–D6 indicators are IFRS 16-oriented and cannot be reused as trade reconciliation rules |
| Multidimensional analytics | Analyse extraction coverage, evidence availability and exception patterns | ReconIQ requires new financial-domain analytics and independent validation |

The most important methodological connection is the separation of evidence availability from answer generation.

However, the proposed four extraction outcomes, financial source hierarchy, field-level provenance structure and settlement-date handling are ReconIQ design decisions. They should not be attributed to the paper.

**9. Limitations**

There are several boundaries to what can legitimately be claimed from the research.

**What the paper establishes**

The paper presents:

- A RoBERTa-based extractive QA framework.
- Explicit answerability classification.
- Conditional masking of span supervision for unanswerable examples.
- IFRS 16-oriented indicator mapping.
- Evaluation using contractual question-answering data and enterprise contract-validation examples.

Its results provide evidence for the proposed methodology within its evaluated contract-intelligence setting.

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**What cannot be claimed**

The paper does not establish that:

1. Its model can reliably extract trade details from broker confirmations or custodian statements.
2. Its answerability classifier can distinguish all four proposed ReconIQ extraction outcomes.
3. Its model can identify corresponding trades across multiple financial systems.
4. Its extraction method can validate settlement dates, prices, quantities or allocations.
5. Its reported metrics predict ReconIQ performance.
6. Its D1–D6 mapping can be directly reused for financial reconciliation.
7. Its framework eliminates the need for financial rules or human review.

**What ReconIQ must independently test**

| **Evaluation area** | **What should be tested** |
|---|---|
| Field extraction | Accuracy of trade date, side, quantity, price, currency, settlement and allocation extraction |
| Answerability classification | Correct classification of answerable and missing evidence |
| Ambiguity detection | Ability to identify conflicting values and multiple interpretations |
| Invalid evidence detection | Performance on unreadable, malformed and incomplete documents |
| Evidence traceability | Whether every extracted value points to the correct source location |
| Abstention quality | Whether unsupported values are correctly left unpopulated |

| | |
|---|---|
| Source generalisation | Performance across different brokers, custodians and document formats |
| Human review | Whether escalated cases contain sufficient evidence for efficient investigation |

ReconIQ should evaluate extraction and answerability separately. For example, a model may extract values accurately when evidence is present but still perform poorly at identifying when information is absent.

It should also measure false extraction: cases in which the system produces a value despite insufficient evidence. In a financial environment, this is an especially important failure mode.

**10. Final deliverable: ReconIQ evidence extraction technical specification**

**ReconIQ — Evidence Extraction Component**

Proposed specification

Functional design

Not paper-validated

**10.1 Purpose**

Extract trade attributes from external financial records while determining whether each requested value is sufficiently supported by the supplied evidence.

**10.2 Inputs**

- Broker confirmations.
- Custodian statements.
- Settlement reports.
- Allocation reports.
- Other approved financial records.
- Requested trade reference and required field list.

**10.3 Supported fields**

- Trade identity and references.
- Security identifier.
- Trade date.
- Buy/sell side.
- Executed quantity.
- Execution price.
- Currency.
- Settlement date and status.
- Account, portfolio and allocation details.

**10.4 Processing requirements**

1. Identify the requested trade attributes.
2. Locate the relevant evidence in the supplied source.
3. Assess whether each attribute is supported by the evidence.
4. Extract only supported values.

5. Preserve the exact evidence quote and source location.
6. Assign one of the four extraction outcomes.
7. Retain confidence and validation information where justified.
8. Escalate ambiguous and invalid evidence.
9. Return missing fields as unknown rather than inferred values.

**10.5 Extraction outcomes**

| **Status** | **System behaviour** |
|---|---|
| ANSWERABLE | Extract and attach evidence |
| NOT_FOUND | Return null and record absence |
| AMBIGUOUS | Preserve competing evidence and escalate |
| INVALID_EVIDENCE | Record failure reason and escalate |

**10.6 Output**

A structured field-level response containing:

- Field name.
- Extracted value.
- Extraction status.
- Exact evidence quote.
- Source document and location.
- Confidence, where meaningful.
- Validation and normalisation information.

**10.7 System constraints**

- No unsupported value generation.
- No silent substitution from internal records.
- No automatic resolution of conflicting evidence.
- No conversion of missing information into agreement.
- No assumption that extraction success means reconciliation success.
- Full source traceability for extracted values.

**10.8 Downstream interface**

The component supplies extracted values, statuses and evidence references to the candidate-matching and reconciliation components.

It does not independently decide whether two trade records represent the same transaction or whether the transaction is financially reconciled.

**Final research positioning**

The strongest connection between ROU-TransQA-v2 and ReconIQ is not the transfer of its RoBERTa model or IFRS 16 indicator mapping. It is the methodological principle that **information extraction must be conditioned on the availability and sufficiency of supporting evidence**.

In the proposed ReconIQ architecture, this becomes an answerability-aware evidence extraction component that extracts supported financial attributes, explicitly represents missing or ambiguous information, preserves source-level traceability and abstains when evidence is insufficient.

This component establishes the reliability of the information entering reconciliation. It does not itself perform trade matching or final financial validation.